



Palaeohydrology of the Lower Cretaceous pre-salt lacustrine system, from rift to post-rift phase, Santos Basin, Brazil

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ABSTRACT

Large oil accumulation discoveries in the last decade in the pre-salt rock succession off the Brazilian coast aroused renewed interest in carbonate reservoirs, prompting several studies aiming at unravelling the conditions of formation of these rocks. Despite that, there are still many gaps in the knowledge concerning environmental conditions of the deposition of the pre-salt carbonates of Santos Basin, located at the southeastern Brazilian margin. In this study, we present and discuss a new suite of geochemical and biostratigraphic data of the lacustrine rift to post-rift succession (broadly correlated with Itapema and Barra Velha Formations) of the Santos Basin, whose deposition occurred in the Aptian. The integration of the petrographic, biostratigraphic and geochemical data serves the ultimate purpose of the study, which is to apply the new radiogenic strontium data as a proxy for the reconstruction of the evolution of the lake hydrology associated with the deposition of the carbonates of the Itapema and Barra Velha Formations. These data are integrated into numerical models, whose results bring new ideas on the general hydrologic conditions of the lake system and on its evolution through time. The mass balance calculations indicate that two end-members alone can account as potential sources of Sr to the palaeolake: a mafic and a felsic component, respectively. The latter likely contributed with significantly higher fluxes, roughly one to two, and up to four, orders of magnitude higher than the former end-member for most of the depositional history in order to generate the $^{87}\text{Sr}/^{86}\text{Sr}$ trends observed in the carbonate rocks. Finally, the results obtained from the integration of the observations with the numerical models point to decreasing fluxes of water with time as contributed from both sources, suggesting that the formation of the deposits became very likely constrained to progressively more evaporitic and alkaline lake waters. These results also open up a new window into the environmental conditions of the interior Gondwana continent immediately prior to the definitive opening of the South Atlantic Ocean.

1. Introduction

The geological and environmental conditions during the Early Cretaceous in the interior of the Gondwana continent, which led to the deposition of thick lacustrine carbonate sequences prior to the definitive opening of the South Atlantic Ocean, are not well constrained. The discovery of a giant oil province composed by pre-salt lacustrine carbonate reservoirs in the last decade, referred to as the pre-salt of Santos Basin, in the Brazilian southeastern margin, aroused renewed interest to understand the conditions in which these unusual rocks formed, and prompted several studies from industry and academia (e.g., Carminatti et al., 2008; Mohriak et al., 2012; Sabato Ceraldi and Green, 2016; Carlotto et al., 2017; Faria et al., 2017; Amthor et al., 2018; Arienti et al., 2018; Fetter et al., 2018; Souza et al., 2018; Tosca et al., 2018;

Tritlla et al., 2018).

Since the first works addressing the Aptian pre-salt rocks from Campos and later from Santos basins several geological models have been proposed (Dias, 2005; Wright, 2012; Dorobek et al., 2012; Harris et al., 2012; Hamon et al., 2012; Rezende and Pope, 2015; Wright and Barnett, 2015; Muniz and Bosence, 2015; Herlinger et al., 2017; Arienti et al., 2018; Fetter et al., 2018; Souza et al., 2018). These have mostly proposed conceptual models or analogues based on texture, sedimentological, stratigraphic or mineralogical aspects that have been detailed so far. In the recent literature, greater emphasis has been given to the Santos Basin uppermost pre-salt succession (Barra Velha Formation), whereas the lowermost coquinas sequence of Campos Basin has received some attention as well (Silva-Telles et al., 1994; Silva-Telles, 1996; Dias, 1998; Carvalho et al., 2000; Dias, 2005; Thompson

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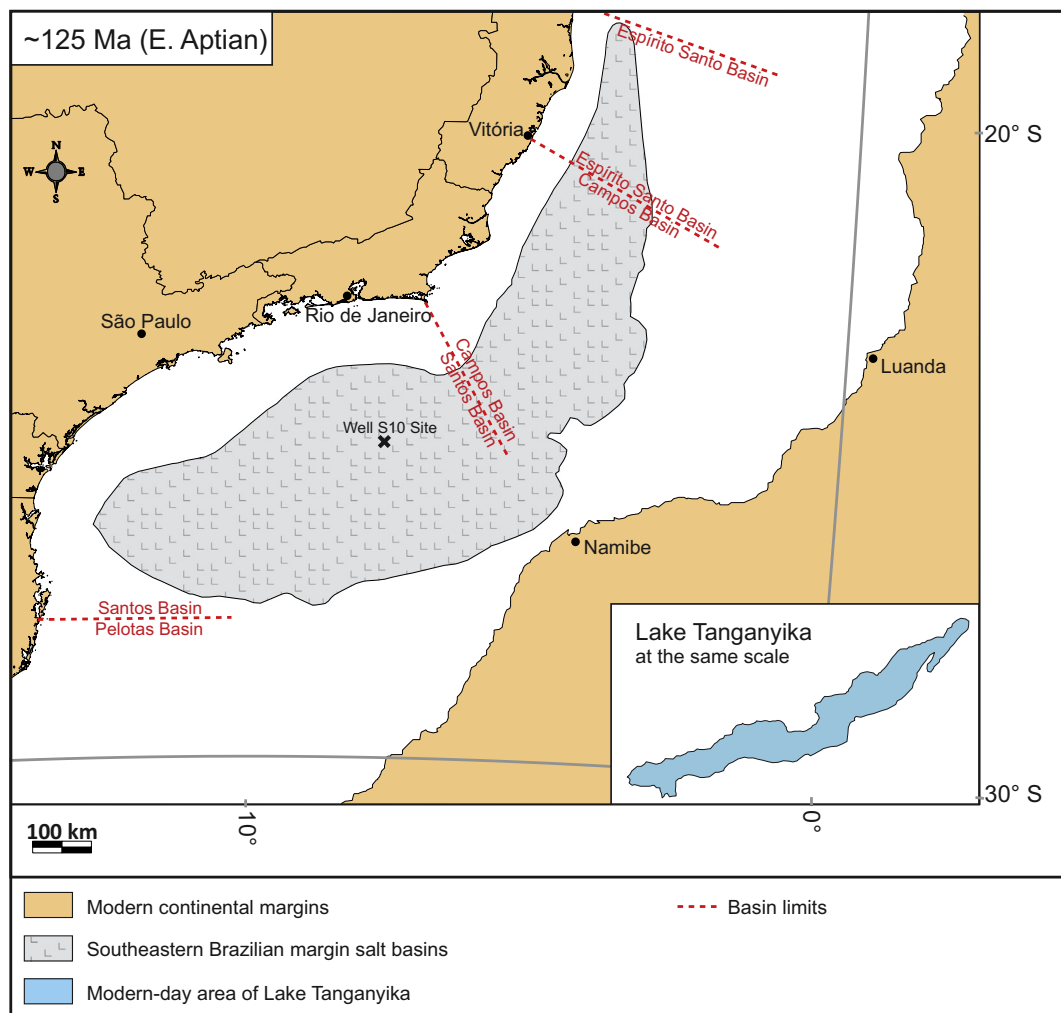


Fig. 1. Palaeoreconstruction of the Central South Atlantic segment during the Aptian (modified from Torsvik et al., 2009), showing the position of offshore Petrobras well S10. Outline of the salt basins of Campos, Santos and Espírito Santo is shown in gray. Only the Brazilian salt margins are portrayed; their African counterpart is omitted. The box in the lower-right corner shows the area corresponding to the Lake Tanganyika (in blue) (the largest modern-day rift lake in the world, after the Baikal in Siberia), rotated and projected to scale with the map, for size comparison. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

et al., 2015; Armelenti et al., 2016). However, to date only a few have addressed in detail geochemical data sets derived from comprehensive rock sequences from these areas, and studies have been mostly concentrated on Campos Basin (e.g., Dias, 1998; Sabato Ceraldi and Green, 2016; Tedeschi, 2017). So far, an integrated approach for the study of the palaeoenvironmental conditions during the deposition of the pre-salt rocks of the Santos Basin, considering biostratigraphy, petrography, sedimentology and geochemistry, and covering a significant stratigraphic record, has not been published yet.

The biostratigraphy of the pre-salt is essentially based on non-marine ostracods, because of the lack of marine fauna in these sequences. Ostracods are aquatic crustaceans, commonly found in different kinds of marine and non-marine water bodies, from the early Cambrian until the Recent (Pokorný, 1998). The ostracods studied herein compose the non-marine group, which experienced an abundance and diversity explosion during the Lower Cretaceous (Whatley, 1992). Among the first biostratigraphers to study the Brazilian Lower Cretaceous biostratigraphy were the Dane Johannes Christian Troelsen and the German Karl Krömmelbein. As a consequence of their work for the oil industry at the time (in the 1960s and 1970s), most of their biostratigraphic charts and reports remain classified and are inaccessible to a significant portion of the scientific community.

The importance of non-marine ostracods as a biostratigraphic tool

has evolved since then. Their relevance to biostratigraphic dating is generally based on the assumption of the broad dispersal of their eggs from one water body to another, as their capability of resisting dryness and wind transport is well documented (Whatley, 1992). Moreover, the improvement of taxonomic studies allowed biostratigraphers nowadays to be capable of identifying which species are suitable for relative dating based on their abundance, and on ease to find them in different Brazilian basins (Do Carmo et al., 2008; Poropat and Colin, 2012b).

Radiogenic strontium isotopes, in turn, have been extensively applied to the study of marine carbonate successions for stratigraphic correlation and palaeoenvironmental reconstruction. Nonetheless, less often, they have also been employed with success to the study of carbonates deposited in continental settings such as lakes (e.g., Rhodes et al., 2002; Gierlowski-Kordesch et al., 2008) and, as put by Doeberbert et al. (2014), strontium isotopes can be exceptionally sensitive indicators of lacustrine evolution. Arguably, they can provide a record of weathering and/or regional drainage patterns, as well as be a reliable tracer of fluid sources.

In this study, we present and discuss a new suite of $^{87}\text{Sr}/^{86}\text{Sr}$ and biostratigraphic data of the lacustrine rift to post-rift (sag) successions of the Santos Basin, comprising essentially carbonate rocks with minor associated detrital siliciclastic content, from offshore Petrobras well S10. These data are interpreted with the support of additional

Fig. 2. Lower Cretaceous stratigraphy of Santos Basin (adapted from [Moreira et al., 2007](#)) and ostracod biozones from Campos Basin (modified from [Silva-Telles, 1992, 1996](#); [Rangel and Carminatti, 2000](#); [Poropat and Colin, 2012b](#)).

2. Geological setting

The figure is based on a palaeogeographic reconstruction for the early Aptian of [Torsvik et al. \(2009\)](#) and shows present-day coastlines for reference. For comparative purposes, the area of modern-day Lake Tanganyika, a rift lake in eastern Africa with approximately 650 km in the main axis and ~50 km wide, is rotated and projected to actual scale in the map, and the outline of the Brazilian salt basins of Espírito Santo,

Above these basalts, [Moreira et al. \(2007\)](#) distinguished and described three other lithostratigraphic units that overlie the basal volcanic sequence and were formed prior to the deposition of the evaporites in the Santos Basin. The basal unit is known as Piçarras Formation, which is characterised, in its proximal part, by alluvial fan conglomerates and coarse to medium-grained sandstones, composed of monocrystalline quartz and feldspar grains, plutonic rock and basalt fragments, and by fine-grained sandstones, siltstones and shales in distal lacustrine portions. The second unit corresponds to the Itapema

Formation, which is generally composed of interbedded bivalve rudstones and grainstones, and black shales. The upper unit is known as the Barra Velha Formation, and has recently been described in greater detail by Arienti et al. (2018) and Souza et al. (2018). This unit is predominantly constituted by shrubby limestones, laminated limestones and dolostones, intraclastic grainstones and hybrid conglomerates and arenites, as well as spherulitic and dolomitic Mg-rich silicate clays.

Above the Barra Velha Fm. lies the evaporitic sequence, constituted by nearly 2 km of salt deposits in some parts of the basin, and corresponding to the Ariri Formation (Moreira et al., 2007). Still above them lies a sequence of marine platform carbonates overlain by thick deposits of coastal, shelf, slope and pelagic sediments, ranging in age from Albian to the Holocene. The evolution of the Santos Basin occurred in such a way that it favoured a progressive burial of the pre-salt rocks, and for this reason, they do not crop out in any portion of the basin. Fig. 2 presents the stratigraphic chart of Santos Basin along with biostratigraphic zones, with focus on the specific interval corresponding to the Early Cretaceous.

The Barra Velha Formation lays unconformably over the Itapema Fm. This limit is usually easily mapped in seismic sections, having a regional expression, and is known in both Campos and Santos basins as the pre-Alagoas unconformity (Winter et al., 2007; Moreira et al., 2007). The age corresponding to the top of Itapema Fm. is not well constrained, but is estimated to be around 120 and 123.1 Ma, according to definitions for the timing of the pre-Alagoas unconformity presented by the latter authors. The Barra Velha Fm. is further subdivided in two stratigraphic sequences; the unconformity bounded lower and upper Barra Velha Formation. The intra-Alagoas unconformity, not always traceable, represents the boundary between these two. The upper Barra Velha Fm. is further limited by the Ariri Fm. deposits above it (Moreira et al., 2007).

The lower and upper Barra Velha Fm. correspond to the upper syn-rift and to the sag phases, respectively (Moreira et al., 2007; Buckley et al., 2015). The facies corresponding to these two sequences have been interpreted in the past as being deposited in a transitional environment, from continental to shallow marine setting (Moreira et al., 2007).

In the majority of the South Atlantic offshore basins the biostratigraphic charts with respect to the Lower Cretaceous interval are based on non-marine ostracods and/or palynomorphs (Schaller, 1969; Viana et al., 1971; Moura, 1988; Silva-Telles, 1992; Poropat and Colin, 2012a, 2012b; Grosdidier et al., 1996). Classically, the biostratigraphic zonation based on non-marine ostracods led to the establishment of six local stages, spanning from the Tithonian to the lower Albian (Moura, 1988), two of which are important to the present study: the Jiquiá and the Alagoas local stages, which span from the Barremian to the lower Albian (Fig. 2). In a broader sense, all of the aforementioned studies converge to the same results, with a few discrepancies appearing only in the detailed approach. Absolute age determinations for this non-marine sequence have been a long-standing problem difficult to tackle. Estimations of the duration of the Jiquiá local Stage, which encompasses the whole of the Itapema Fm., have varied from 2 to 5 Myr (Silva-Telles, 1996). Estimations for the Alagoas local Stage, comprising the entirety of the Barra Velha and Ariri Formations, have been an even worse problem, because of inconsistencies with respect to the upper limit of its type section, its lower faunistic diversity of ostracods and scarcity of any other fossil group useful for the purposes of relative dating.

In a recent contribution though, Tedeschi et al. (2017) presented new evidence in which the evaporites above this entire sequence may be correlated to the early Aptian Oceanic Anoxic Event 1a (OAE 1a), around ~120 or ~125 Ma (according to Malinverno et al., 2010 and Ogg et al., 2012, respectively). Thus, previous estimates of ages for the sequences below the evaporites are unquestionably pending thorough revision as well.

In addition to the depositional history of the pre-salt succession, a

long-standing debate on the occurrence of the first marine incursions during the opening of the South Atlantic has also involved these deposits. Silva-Telles et al. (1994) have attributed marine incursions into Campos Basin already during Jiquiá times, which corresponds to the deposition of the coquinas from the Lagoa Feia Group (the counterpart of Itapema Fm. in the Santos Basin), but this has been contested by Thompson et al. (2015). Dias (2005) attributed a marine deposition to the carbonates of the Macabú Formation (Lagoa Feia Group, Winter et al., 2007), Campos Basin, which is adjacent to Santos Basin, possess the same facies association and is considered to be time equivalent to the Barra Velha Formation. Moreira et al. (2007) and Gomes et al. (2009) suggested that the latter formation was deposited in a transitional environment, under marine influence. In turn, Wright and Barnett (2015) also added to the controversy regarding the depositional history of these carbonates, discarding the hypothesis of marine origin and favouring the deposition of the Barra Velha carbonates in an alkaline lacustrine environment, based on sedimentological and mineralogical descriptions. Recently, Tedeschi (2017) has provided strong evidence in support of this latter view, based on $^{87}\text{Sr}/^{86}\text{Sr}$ data from carbonates of the Campos and Santos basins. Finally, Souza et al. (2018) have presented a new petrologic study in which the authors interpret many of the Barra Velha carbonate rocks as travertine-related deposits, derived from a significant contemporaneous hydrothermal component of this system.

Our study is focused on the Itapema and Barra Velha Formations, correlated to the Jiquiá and Alagoas local stages based on ostracods, respectively, covering the syn-rift to post-rift (sag) sequences. These in turn have been traditionally considered to range approximately from late Barremian to late Aptian (Moreira et al., 2007; Poropat and Colin, 2012a; Buckley et al., 2015; Thompson et al., 2015; but see also Tedeschi et al., 2017). The studied rocks are predominantly composed of carbonates, with variable proportion of silicate minerals.

3. Methods

3.1. Sampling and material

The data used in this study were all acquired from well S10 (Fig. 1). The studied interval encompasses > 800 m of rock record, whose present-day depth is nearly 5000 m below sea level. The samples for the geochemical study comprise recovered sidewall cores roughly 3 cm in length per two and a half centimetres in diameter in average. Usually only a small fraction of a few grams is harnessed for chemical studies and biostratigraphic analyses, and thus allowed to be completely consumed.

3.2. Cleaning protocol for geochemical analyses

Prior to the geochemical analyses, all samples were submitted to oil and salt extraction. This was conducted in two steps using Soxhlet apparatuses: (1) using an azeotrope mixture consisting of methanol and chloroform (30:70 v/v), until no fluorescence under 365 nm wavelength UV light was observed, which was followed by (2) extraction with methanol and using 10% AgNO_3 for checking salt removal. The absence of AgCl precipitates indicated complete removal of the latter.

After drying all available samples were crushed and ground to a size below 80-mesh powder using agate pestle and mortar, and were homogenised. Aliquots were then taken for the geochemical analyses.

3.3. Biostratigraphy

The non-marine ostracod analyses were carried out on 123 composite cuttings, each individual sample comprising material collected within every 6 m, and on 60 additional cuttings collected in three-metre intervals along the intermediate portion of the well. Additionally, 36 sidewall core samples, unevenly distributed along the well, were also

used for this purpose. The samples were processed according to the standard methodology, starting by soaking 60 g of sediments (in the case of cuttings) in a 30% hydrogen peroxide solution until the complete oxidation of organic matter. Next steps included sample washing on a 0.125 mm sieve and handpicking of total specimens. The identification of the picked specimens was done using a stereomicroscope with reflected light.

3.4. Petrography

Altogether, 151 sidewall core samples were analysed, described and classified under a transmitted polarised light optical microscope, considering texture, fabric and compositional aspects. Hence, modal compositions of primary and diagenetic minerals and paragenetic interpretations were obtained through systematic qualitative petrography in each thin section prepared from samples impregnated with blue epoxy resin. Staining with hydrochloric solution of alizarin Red-S and potassium ferrocyanide was performed in order to differentiate the carbonate mineral phases (cf. Friedman, 1959). Scanning electron microscopy analyses were carried out on carbon-coated selected thin sections in a JEOL JSM-6690LV microscope with support from an Oxford-Inca energy dispersive spectrometer (EDS) for a finer characterisation of the elemental composition of some constituents.

3.5. Elemental composition

A total of 194 sidewall core samples were analysed for determination of nine selected elements: Sr, Ca, Mg, Mn, Fe, K, Rb, Al and Si. From that total, 192 intact sidewall core samples were analysed on their flat surfaces using an Olympus Delta Premium 6000 handheld emission dispersive X-ray fluorescence device (XRF) after the cleaning protocol. This instrument is equipped with a Rh anode working at 4 W as an excitation source and a 40 kV tube. Samples were analysed in a workstation for operator protection under atmospheric environment, in both Soil and Mining Plus modes.

In addition, 151 sidewall core samples had enough material to follow grinding and powdering, after which they were analysed by ICP-MS and ICP-OES. Determination of major elements Ca, Si, Al, K and Rb was carried out on powdered samples fused in lithium metaborate and analysed after digestion in HCl for determination of whole rock composition. The first four elements were analysed in an Optima 7300 DV Perkin Elmer inductively coupled plasma optical emission spectrometer (ICP-OES), whereas Rb was determined in an Elan DRC II Perkin Elmer ICP-MS. The elements Mg, Fe, Mn, and Sr were also determined in the ICP-MS, after digestion of the powders in aqua regia solution (3:1 HCl:HNO₃ v/v) for extraction of elements bound essentially to the carbonate phase.

Reproducibility and precision of the methods were based on the analysis of Certified Reference Materials and laboratory duplicate samples. Reference materials IPT-122 (dolomitic limestone), IPT-44 (limestone), NIST-1d (argillaceous limestone), NIST-88b (dolomitic limestone), and BRP-1 (basalt) were used to calibrate XRF analyses and verify both reproducibility and precision of the methods above. A total of 149 samples overlap between XRF and ICP methods, so that quality control could be performed in order to verify discrepancies that could arise from the use of results derived from different methods. The analytical data can be consulted in the Supplementary material which is available upon request.

3.6. Strontium-isotope analyses

194 rock samples were analysed for ⁸⁷Sr/⁸⁶Sr ratio. For this purpose, 1 mg of carbonate powder was digested in weak acetic acid to dissolve only the carbonate fraction and avoid leaching of ⁸⁷Sr and Rb from non-carbonate phases that might be present, which was followed by separation of Sr from the eluent in Teflon chromatographic columns

filled with SrSpec (100–150 µm) resin, after the first leaching. ⁸⁷Sr/⁸⁶Sr ratios were determined using a Triton Plus thermal ionization mass spectrometer in static multicollector mode at the Geochronology Laboratory of the University of Brasília. Analyses were calibrated with measurements of the NBS-987 standard (SrCO₃), yielding an average value of 0.710280 ± 0.000010 (2SD). All ⁸⁷Sr/⁸⁶Sr data were corrected for fractionation assuming a ⁸⁸Sr/⁸⁶Sr ratio = 0.1194.

3.7. Statistical analysis

For statistical treatment all elements were converted to the same unit, ppm. Those originally reported as oxides were also converted to element concentration. A composite table was built to include the additional set of samples that had results originated only from XRF analysis and to report all data together for the following steps necessary to statistical assessment (see Supplementary material; Supplementary material is available upon request).

Prior to statistical analysis, elements were evaluated to account for the number of undetected values. The composite data of potassium (ICP-OES + handheld XRF data) represented the largest set of samples with lower than limit of detection results (amounting 16.8% samples below LOD for both ICP-OES and XRF, where K₂O < 0.01 and < 0.02%, respectively). These “zero” results were replaced by a random value between zero and the respective limit of detection and the same followed for the other elements that were in similar situation. Once the data were properly organised they were ready for statistical analysis.

In order to better assess the elemental data, in addition to the traditional evaluation of bivariate plots between isotope and element ratios, as in some cases used further below, Principal Component Analysis (PCA) was employed to evaluate the relationship among all the elements. The need for caution when drawing inferences from simple correlations between single elements or their ratios to some common denominator is a subject already well documented in geochemistry and chemometrics more generally (e.g., Aitchison, 1986; Rock, 1988; Rollinson, 1993; Varmuza and Filzmoser, 2009; Filzmoser et al., 2009). Compositional data carry the intrinsic characteristic that no individual component can be interpreted isolated from each other, derived from the fact that:

$$x_1 + x_2 + \dots + x_D = 1 \quad (1)$$

where the terms in Eq. (1) correspond to individual element concentrations.

The set of compositional data as a whole comprises a vector, or a matrix, that is subjected to the effects of unit-sum constraint, or closure, where one single sample is represented in the following way (Aitchison, 1986):

$$\mathbf{x}^{(D)} = (x_1, x_2, \dots, x_D) \quad (2)$$

This constraint can eventually lead to forced correlations between variables. In order to avoid this problem, before multivariate statistical analysis the compositional data were first treated following the mathematical procedure of centered log-ratio transformation (Aitchison, 1986; Filzmoser et al., 2009; Montero-Serrano et al., 2010), which is defined as:

$$clr(\mathbf{x}) = (y_1, \dots, y_D) = \left(\ln \frac{x_1}{\sqrt[D]{\prod_{i=1}^D x_i}}, \dots, \ln \frac{x_D}{\sqrt[D]{\prod_{i=1}^D x_i}} \right) \quad (3)$$

As will be seen later, this transformation allows good evaluation of the results of Principal Component Analysis at the level of exploratory data analysis, significantly reducing bias that might otherwise occur. These results allow to address important aspects with respect to the behaviour of the elements and to support additional inferences concerning palaeoenvironmental or diagenetic conditions associated with the samples.

3.8. Box-model

In order to try to reconstruct the main fluxes that may have contributed to the original lake, we used a simple box model derived from the equation of Richter et al. (1992), which is slightly adapted to our case:

$$N \frac{dR_{lw}}{dt} = F_m(R_m - R_{lw}) + F_f(R_f - R_{lw}) - F_s(R_s - R_{lw}) \quad (4)$$

where N stands for the total number of moles of Sr in the lake water, R_{lw} is the $^{87}\text{Sr}/^{86}\text{Sr}$ of the lake water, F_m is the flux of Sr into the lake derived from a mafic source end-member, which has an average $^{87}\text{Sr}/^{86}\text{Sr}$ of R_m , and F_f is the flux of Sr derived from a felsic source end-member, with average $^{87}\text{Sr}/^{86}\text{Sr}$ of R_f . F_s is the flux to the sink, represented by precipitation of carbonate sediment, and R_s corresponds to the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of the sink, which is assumed to be equivalent to that of the lake water. Hence, the last term of the equation cancels out. The remaining parameters and the numerical values used in the model will be addressed in detail in the discussions further below.

Finally, the same modelling conditions were applied to two different age scenarios for the deposition of the Barra Velha Fm., because of the uncertainty regarding its timing and of the relevance to test how different depositional rates would affect our initial premises. These scenarios are henceforth called scenario 1 and scenario 2.

4. Results and discussion

4.1. Biostratigraphy and age estimation

The collected ostracods span a biostratigraphic record comprised of five bioevents for the Jiquiá local Stage, relative to species extinctions, and three bioevents for the Alagoas local Stage (Fig. 3), similarly characterised by extinctions of species. The associations found in the

well comprise the biozones OS-1010 to RT-011 described in the Campos Basin by Silva-Telles (1992, 1996) and Poropat and Colin (2012a), suggesting a relatively continuous record. Considering only the Jiquiá interval, the five bioevents shown in Fig. 3 exhibit a decrease in the number of genera, from three to one, that seem to be more sensible to the degree of salinity and/or alkalinity towards the top of the well. In the lower portion of the well the recovered Jiquiá genera *Reconcavona* and *Petrobrasia* extinguish and the same happens in the record of the upper Jiquiá to *Hourcquia* and *Ilyocypris* genera. The Jiquiá-Alagoas transition occurs along a sparsely fossiliferous interval with badly preserved shells. The ostracods retrieved from the Alagoas local Stage belong to the same genus (*Pattersonocypris*), but the abundance and richness are remarkable at the base and at the middle of the studied section, both decreasing to the top. The upper part of the Alagoas becomes barren in ostracods until the base of the evaporites. Recovered species show that in the studied well, from lower Jiquiá to the upper Alagoas, the ostracod fauna experienced a decrease in ornamented and robust shells as the extinction of these genera gives place to thinly skinned carapaces. This observation, in turn, indicates that the environmental conditions of the lake in which they lived became progressively more stressing with time (Bate, 1999).

Regardless of the uncertainties concerning the exact timing of the deposition of the pre-salt rocks, as introduced above, the nearly complete ostracod record suggests significant continuity in the deposition of the sequence. The pre-Alagoas unconformity represents the main hiatus, and the span of time comprised in this gap is not clear, because of the absence of marine fossils, and consequently, the impossibility of direct calibration of the lacustrine sequence as a whole against the formal international chronostratigraphy, based on marine sections. Nonetheless, it has been traditionally accepted to be an erosion/non-deposition episode around 120 and 123.1 Ma, according to interpretations published by Winter et al. (2007) and Moreira et al. (2007) and

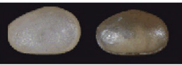
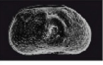
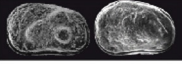
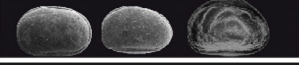
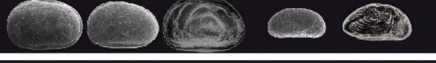
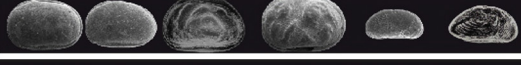
Age (Ma)	Bioevents	Genera Stratigraphic Distribution	Retrieved ostracods
113/120* Ma Aptian (Alagoas Local Stage)	BE-8	<i>Pattersonocypris</i>	
	BE-7		
	BE-6	<i>Ilyocypris</i>	
123 Ma Aptian (Jiquiá Local Stage)	BE-5	<i>Hourcquia</i>	
	BE-4		
	BE-3	<i>Reconcavona</i> and <i>Petrobrasia</i>	
	BE-2		
	BE-1		
126 Ma			

Fig. 3. Retrieved ostracods in samples from the well presented in this study, depicting a biostratigraphic record of five bioevents (extinctions) for the Jiquiá local Stage and three bioevents (extinctions) attributed to the Alagoas local Stage, covering the full range of biozones expected for these intervals, according to Fig. 2 and to Silva-Telles (1992, 1996), Rangel and Carminatti (2000) and Poropat and Colin (2012b). The pre-Alagoas unconformity is implied in this figure, and is assumed in the present study to last between ca. 1–2 Myr, based on inferences reported by Winter et al. (2007) and Moreira et al. (2007). Two different ages are depicted as possible upper boundaries for the studied interval: 113 Ma, based on traditionally published stratigraphic charts (e.g., Moreira et al., 2007) and 120 Ma, based on more recent data (cf. Tedeschi et al., 2017), where the asterisk refers to the 120 Ma upper boundary. See the text for details.

the biozone calibration attempt made by Poropat and Colin (2012a), as previously shown.

Moreover, as introduced above, the age of the bottom of the evaporites has been generally accepted to be around 112 or 113 Ma until very recently (cf. Dias et al., 1994; Dias, 1998; Dias, 2005; Davison, 2007; Karner and Gambôa, 2007). The latter age has been taken further to constrain the top of the Barra Velha Formation, according to Moreira et al. (2007). Nonetheless, Tedeschi et al. (2017) suggest that these assumptions are probably wrong, and that the timing of the deposition of the evaporites very likely occurred during OAE 1a. This would place the top of the Barra Velha Formation at ~120 or ~125 Ma.

Consequently, we considered the two different age scenarios in our models with reference to time below. The first one (scenario 1) assumes the former age estimate and hence a considerably stretched interval of time for the deposition of the Barra Velha carbonates, and the other (scenario 2) assumes the bottom of the evaporite sequence to be 120 Ma old, which results in a comparatively more compressed time period for the deposition of the Barra Velha Formation.

To avoid inconsistencies we assume the age of ~123 Ma to limit the top of the Itapema Fm., for both scenarios above. The age of the bottom of the entire sequence is constrained by the correlation of the Camboriú basalts with the Cabiúnas basalts of Campos Basin, whose age range between 134 and 122 Ma (Mizusaki et al., 1992; Winter et al., 2007; Moreira et al., 2007). We fix the bottom of the Itapema Fm. at ~126 Ma in the two scenarios for the purposes of modelling. Thus, any changes that could be imparted to our calculations would be exclusively related to the extent of time of the deposition of the Barra Velha carbonates.

Scenario 1 results in a minimum sedimentation rate of approximately 131 m/Myr and 54 m/Myr for the Jiquiá and the Alagoas, respectively. Both are close to present-day sedimentation rates found in lakes belonging to rift settings, such as the Baikal and the Tanganyika (Casanova, 1994; Talbot and Allen, 1996). These sedimentation rates for carbonates are possibly somewhat underestimated, because they do not take into account the effect of compaction. Nevertheless, in a recent forward modelling study Faria et al. (2017) estimated a depositional rate of 80 m/Myr for the Barra Velha carbonates which, although higher than the value used here, lies in close range with it.

Scenario 2, on the other hand, results in a sedimentation rate of 131 m/Myr and 210 m/Myr for the Jiquiá and the Alagoas, respectively. Although the sedimentation rate of the Barra Velha carbonates is nearly four-fold greater in this scenario, this rate of deposition is actually within the lower range of deposition of some modern lacustrine carbonates and travertines, for instance (e.g., Lamb et al., 2007; Pentecost, 2005; Fouke, 2011).

4.2. Petrography

The lower part of the studied section, the Itapema Formation, contains bivalve rudstones and grainstones (“coquinas”), arenites composed of Mg-silicate ooids, peloids and intraclasts and bioclasts, as well as subordinated fine-grained laminated mudrocks and “marls” with variable organic matter content (Fig. 4E–H). In some samples, authigenic quartz occurs either cementing or replacing both primary and other diagenetic constituents, mainly Mg-silicates. Dolomite also occur either as cement or as replacement in some samples. Ostracods are common amidst these rocks. In some samples, phosphate bioclasts of up to a few millimetres, interpreted as being the remnants of vertebrate organisms, are also found. From the palaeontological point of view, the fossil record observed throughout the whole interval indicates that significantly arid conditions predate the Alagoas time, since no evidence of vegetation of any kind is found. Nevertheless, the Jiquiá was probably relatively more humid than the Alagoas, as indicated by the ostracod fauna discussed above and by further geochemical data.

The upper interval, represented by the Barra Velha Formation (Fig. 4A–D), belongs to the Alagoas local stage, and is predominantly composed of syngenetic fascicular-optic calcite aggregates, with

characteristic crystal-shrub and laminated (palisade) fabrics (Wright and Barnett, 2015; Herlinger et al., 2017; Souza et al., 2018), with the predominance of the former. There are also occurrences of spherulitic and fascicular-optic calcite aggregates partially replacing and displacing Mg-silicate laminations and particles, suggesting an eogenetic origin of these components. Mg-silicate constituents are commonly dissolved, silicified or dolomitized. Intraclastic grainstones are recurrent and mainly composed of fragments of either fascicular or spherulitic aggregates, recording successive events of hydrodynamic reworking inside the lake (not shown). The upper portion is largely devoid of significant detrital material, with subordinated presence of silt-sized quartz, feldspar and mica grains and detrital clay. Bioclastic material is scarce and represented by rare ostracod shells. These sedimentological evidences suggest that fluvial input was likely greater or more significant during the time of deposition of the Itapema Formation, in the Jiquiá.

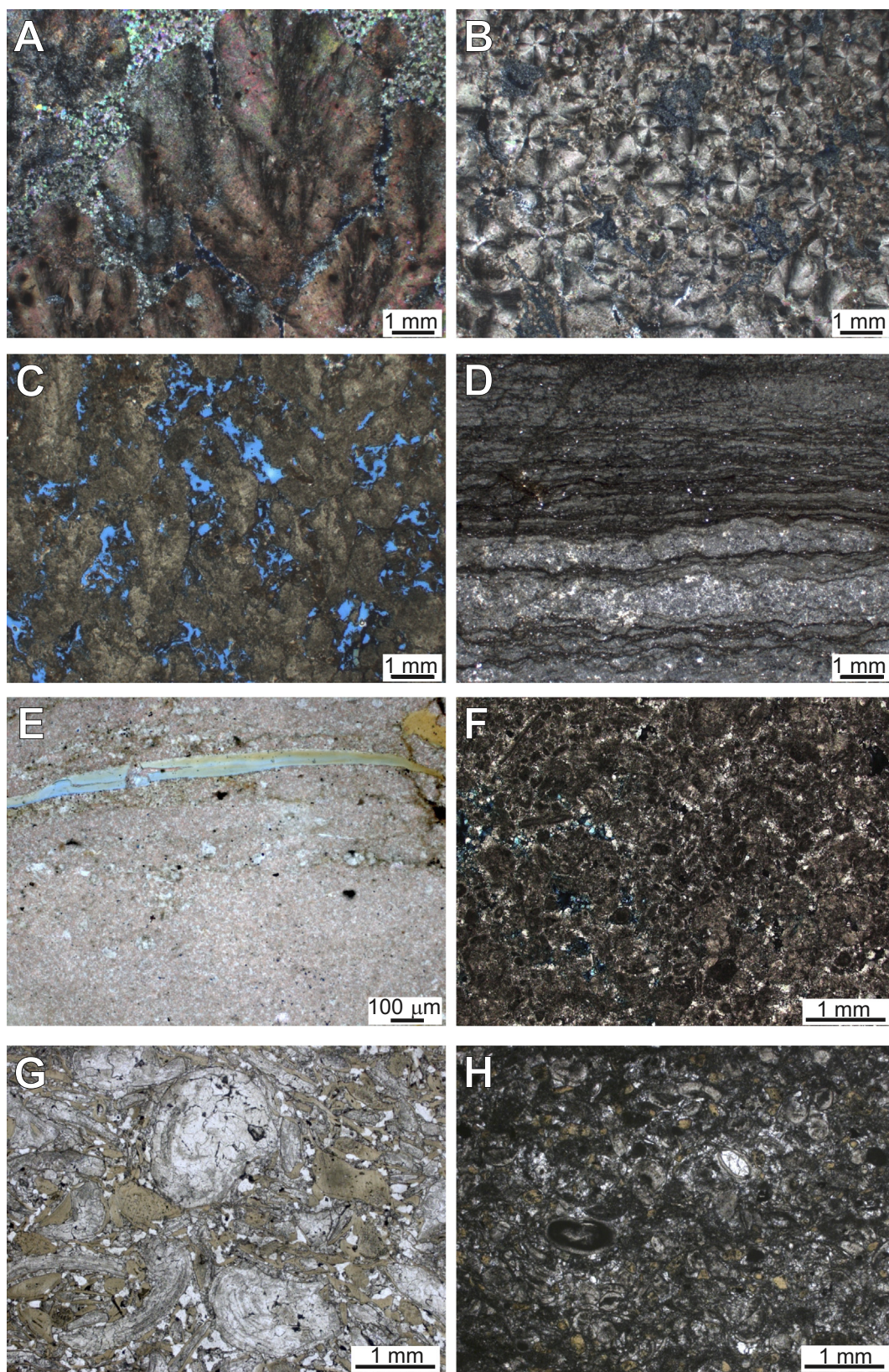
Fig. 4 illustrates some of the most common primary and diagenetic constituents found in the studied well. Mg-silicate constituents (e.g., kerolite and stevensite) are labile and more susceptible to be either dissolved or replaced by calcite, silica and dolomite, being frequently underestimated in the past. Just recently, few authors highlighted such features both in rift and sag intervals of the Campos and Santos basins (Goldberg et al., 2014; Armelenti et al., 2016; Tosca and Wright, 2015; Souza et al., 2018).

Recently, Alvarenga et al. (2016) documented for the first time occurrences of hydrothermal vents associated to the rift-phase Lagoa Feia Group of Campos Basin. Winter et al. (2007) had previously interpreted that this group contains mineralogical associations, such as talc and stevensite that the authors interpreted as products of hydrothermal activity within alkaline volcanic lake(s) in Campos Basin. Hence, hydrothermal activity was likely also common in Santos Basin, as also suggested by Souza et al. (2018). In spite of that, seepage unrelated to hydrothermalism could also have been another source of ions to the lake. Discharge of groundwater as cold seeps or springs can be a significant component of recharge in many lakes, as documented, for instance, in the Ethiopian plateau and in the Great Salt Lake (Kebede et al., 2008; Hart et al., 2004).

4.3. Geochemical data

The handheld XRF has its limitations, and the analysis of unprepared raw samples increases the chances of occurrence of greater dispersity of X-ray energy. In addition, the nature of the heterogeneous rock sample itself prevents optimised analytical performance. Although not ideal, these were practical limitations that had to be overcome in order to provide additional insight into our conceptual model and test it more critically and quantitatively. Overall, the analytical results are fairly good, as shown in Fig. 5. Naturally, the correlations between ICP-OES and handheld XRF data and ICP-MS and handheld XRF data in these cases are not exactly perfect, but the results show statistically significant levels of agreement ($p < .01$, $df = 147$; and $p < .01$, $df = 146$, respectively). This supports the use of XRF data as complementary to the whole set of analysed samples, once it is done with caution and parsimony.

In terms of geochemical composition, the graphics in Fig. 6 also distinguish clearly the two main intervals, Jiquiá and Alagoas. The figure compiles scatterplot data of $^{87}\text{Sr}/^{86}\text{Sr}$ ratio vs. $1/[\text{Sr}]$, Rb/Sr , Sr/Ca , Mn/Sr and Mg/Ca , providing another indication that different sources contributed in variable proportions with time to the overall isotopic composition and Sr concentration of lake waters. One mixing line is suggested from Fig. 6A and B, the $^{87}\text{Sr}/^{86}\text{Sr}$ vs. $1/[\text{Sr}]$ and $^{87}\text{Sr}/^{86}\text{Sr}$ vs. Rb/Sr , respectively, although it cannot be discarded that this effect is produced as a result of greater siliciclastic proportion at the base and its subsequent decrease upwards in the sequence. However, Sr concentrations in most samples were determined after selective digestion, and not from bulk rock analysis, as explained above, derived



(caption on next page)

essentially from carbonates. Thus, it seems likely that the trend observed actually reflects a mixture of fluids from different sources, one more impoverished in Sr and another, which was not. In both cases, though, it seems reasonable to assume that the Jiquiá was more

drainage influenced than the Alagoas, with lower Sr concentration derived from somewhat more diluted waters. Another trend in Fig. 6A, represented by the samples within the ellipse area, likely correspond to recrystallized phases that may have lost some Sr content, but probably

Fig. 4. Photomicrographs of selected samples, representative of the main constituents of Itapema and Barra Velha Formations, observed in the studied well. A–C images correspond to samples from the Alagoas (Barra Velha Fm.) sequence. Image D is from the Alagoas, but similar facies also occur in the Jiquiá (Itapem Fm.). E–H images correspond to samples from the Jiquiá. Images A and C correspond to facies 1 of Wright and Barnett (2015), B to facies 2 and D to facies 3. A: Shrub-like fascicular-optical calcite aggregates cemented by microquartz intercalated with dolomitized syngenetic matrix (XP: crossed polarizers). Calcite aggregates, actually non-magnesian, are fully recrystallized although preserving the original fibrous crystal fabric. B: Calcite spherulites in silicified syngenetic Mg-silicate matrix (XP). Mg-silicate laminations are partially preserved and occur as remnants. Dolomite occurs as rims covering calcite spherulites. C: Shrub calcite aggregates (/P: uncrossed polarizers). Growth-framework porosity (blue-dye stained resin) was enlarged by partial dissolution. D: Laminated limestone: intercalation between thin organic-rich mud and organic-poor carbonate (/P). See micrograph E for detail. E: Phosphate bioclast in undifferentiated mud matrix replaced by microcrystalline calcite (alizarin-dye stained), in laminated limestone (/P). Note streaks of organic matter and Mg-phyllsilicate remnants, and silt-sized detrital grains dispersed in diagenetic calcite matrix. Calcite is here interpreted as diagenetic since the characteristic fabric of micrite neomorphism was not identified and due to the presence of ‘ghosts’ of original particulate deposit. F: Arenite composed of carbonate and Mg-silicate pelloids, partially cemented by dolomite (XP). G: Arenite composed of neomorphosed bivalve bioclasts and Mg-silicate ooids. Packing is tight due to plastic deformation of the ooids (/P). H: Arenite composed of ostracod bioclasts and Mg-silicate pelloids (/P). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

did not have their isotopic composition significantly affected.

From both Figs. 6 and 7 (below) it can be observed that the carbonates from the studied sequence as a whole have characteristically radiogenic signals, which are significantly higher than Aptian marine values, these in turn being lower than ~0.7075 (e.g., Bralower et al., 1997; Bodin et al., 2015). Such high values indicate a significant continental influence, strongly supporting the deposition of the carbonates in a lacustrine environment. Moreover, a few additional distinctions can be made. Samples from Jiquiá are predominantly limited to a range of lower $^{87}\text{Sr}/^{86}\text{Sr}$ values, below 0.7115 in general, whereas the opposite occur for samples from the Alagoas, which are roughly between 0.714 and 0.712. It is observed that the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio generally increases with increasing carbonate content towards the top of the analysed section. The Rb/Sr ratio as measured in the samples is usually well below 0.001 in roughly 60% of the cases (114 samples), becoming consistently lower upwards (Fig. 6B). These observations also strongly indicate that contamination due to leachate contaminants in our samples should not be considered as a reason for the high $^{87}\text{Sr}/^{86}\text{Sr}$ ratio observed in the upper part of the section, since they become more radiogenic upwards with increasing carbonate prevalence, and therefore provide a good indication of actual lake water evolution. Fig. 7 also shows the concentration profiles of Ca, Sr, Al, Si and Mg along with that of $^{87}\text{Sr}/^{86}\text{Sr}$. From these profiles, one can see that the contribution from detrital material is relatively low, but was somewhat higher at the

bottom of the well (as indicated by the Al and Si profiles where they show the same trends and correlate), whereas the carbonates prevail along most of the studied interval. It is also possible to see that these carbonates have a significantly high content of Sr, and that there is indeed a difference in Sr concentration between the Jiquiá and the Alagoas local stages, in which the latter becomes more enriched in this element. Authigenic Mg-silicates and silica, eventually, are another relatively important secondary component of the system (indicated by the Si profile, uncorrelated with Al, and the Mg profile), often occurring together with the carbonates in the Barra Velha and Itapema Fm., as described above. Moreover, dolomite also accounts for a significant portion of Mg present in the samples throughout the well.

These figures suggest that the Alagoas derives its relatively higher strontium concentration from evaporation and concentration of lake waters, as no other mixing line seems evident. The heavier $^{87}\text{Sr}/^{86}\text{Sr}$ end-member probably reflects a contribution of initially less Sr-enriched but highly evaporated waters, as we will discuss further below (Fig. 6D). It could also be argued, on the other hand, that simply a different Sr/Ca partition between solid and fluid could be the reason for such distinction. But other aspects related to the chemical composition of the carbonate samples and their mineralogy do not seem to favour this hypothesis. These are discussed in more detail below, with particular emphasis on the Mg/Ca ratio. Also, the Sr/Ca partition coefficient (K_d) may vary as a function of the Sr/Ca ratio in the liquid, temperature

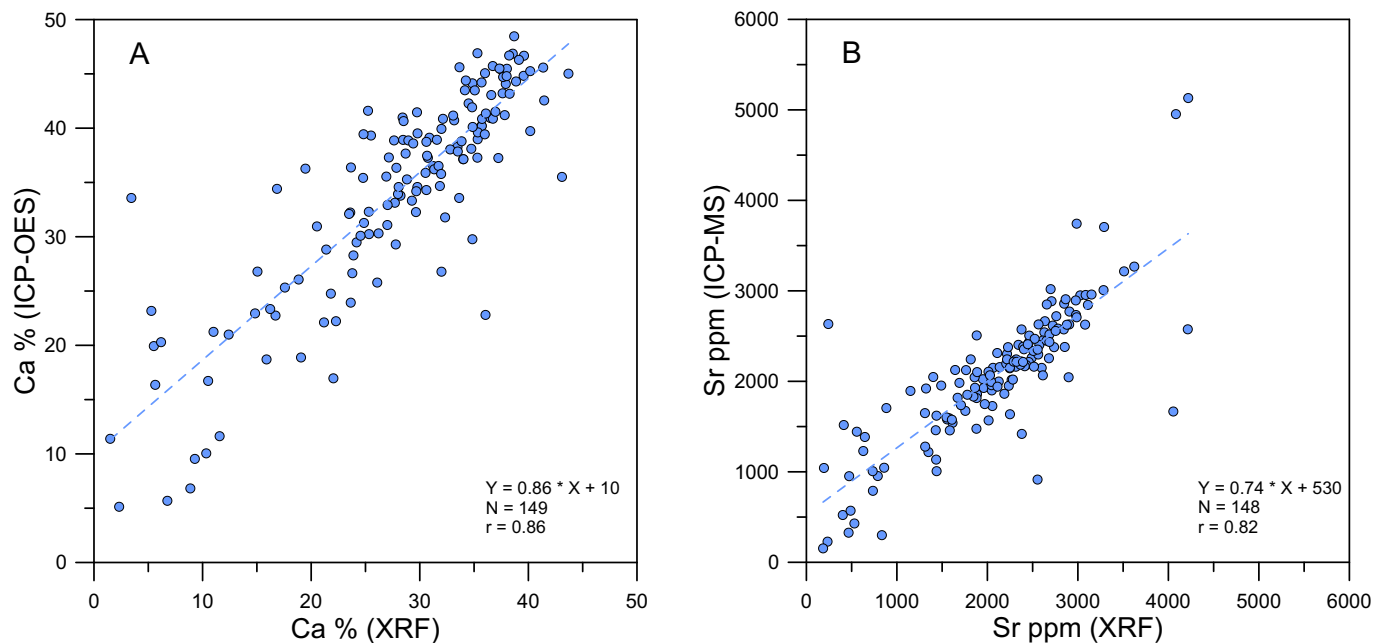
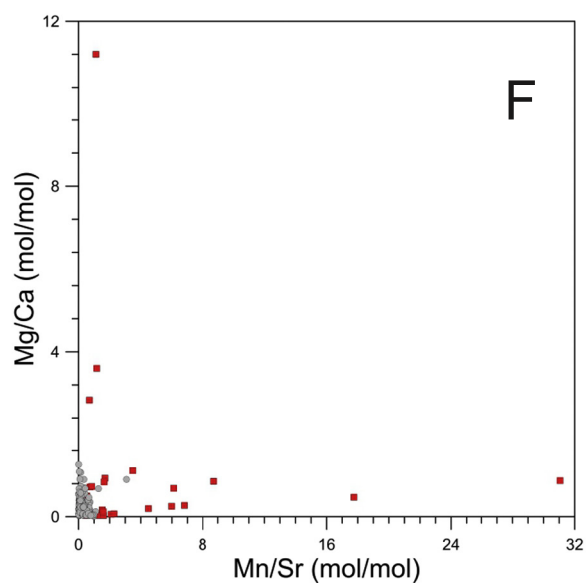
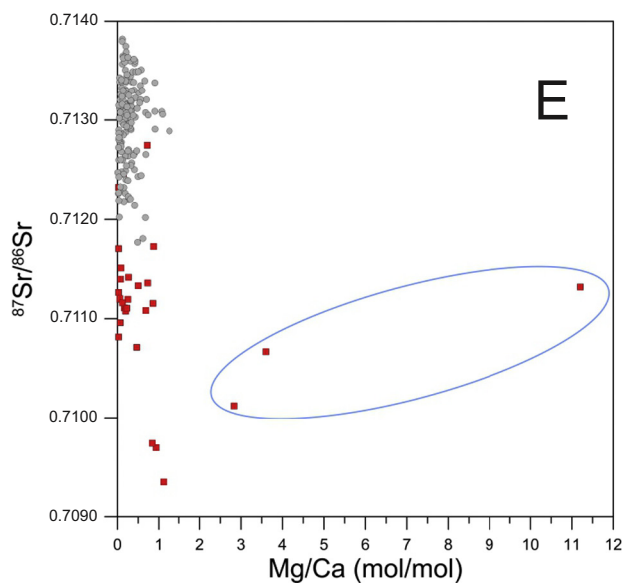
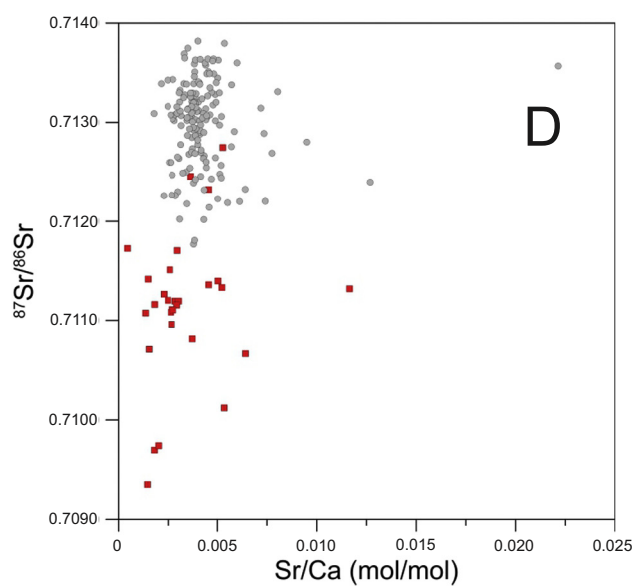
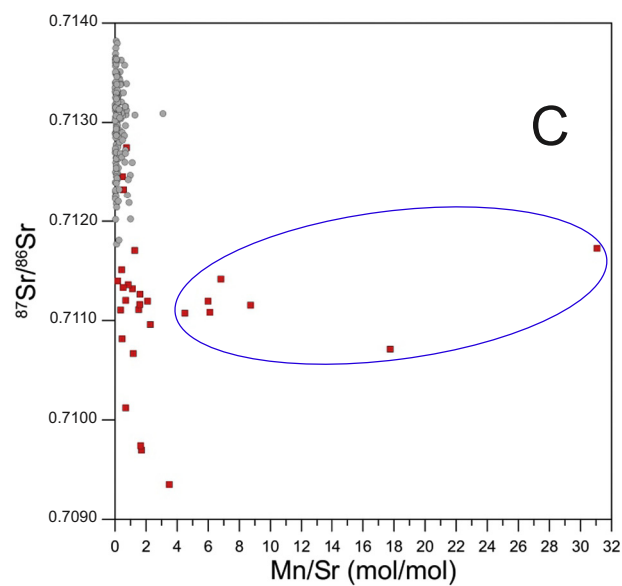
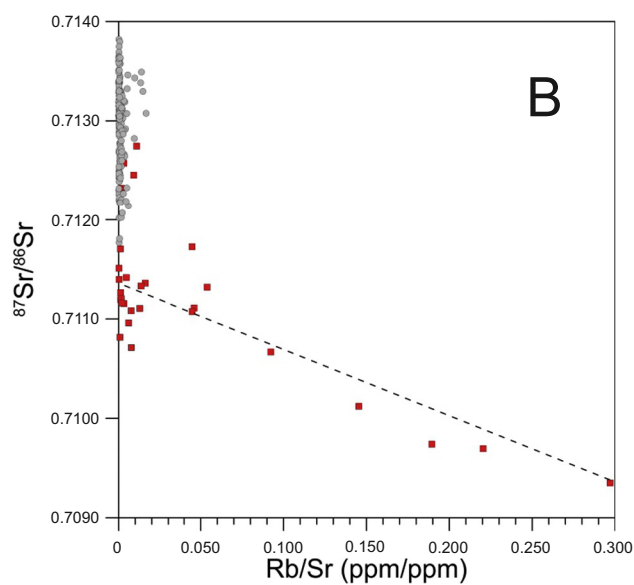
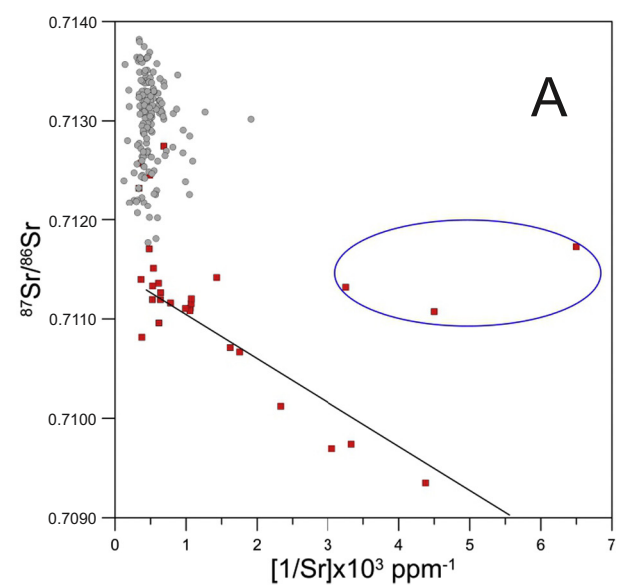


Fig. 5. Crossplots between ICP-OES vs. handheld XRF and ICP-MS vs. handheld XRF data, for elements (A) CaO (%) and (B) Sr (ppm), totalling 149 and 148 sidewall core samples, respectively. One outlier was removed from the Sr graph. Dashed lines: linear regression; r is the Pearson correlation coefficient ($p < .01$, $df = 147$ and $p < .01$, $df = 146$, respectively), and the equations of linear regression are also displayed.



(caption on next page)

Fig. 6. Scatter plots of $^{87}\text{Sr}/^{86}\text{Sr}$ vs. $1/[\text{Sr}] \times 10^3$ (A), Rb/Sr (B), Mn/Sr (C), Sr/Ca (D), Mg/Ca (E), and Mg/Ca vs. Mn/Sr (F). Red squares represent samples belonging to Jiquiá local stage and gray circles are samples from Alagoas local stage. Lines suggest trends of mixing of fluids, although end-member compositions cannot be inferred clearly. Ellipses indicate samples that were likely more significantly modified after burial, although their Sr isotopic ratio apparently was not greatly altered. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and kinetics of precipitation, although it does not seem to vary appreciably in higher temperature ranges of deep diagenesis (Jacobson and Uzdowski, 1976; Gabitov and Watson, 2006). Environmental temperature change is not expected to have been significant, for at that time span the lake system was already within the tropical range, and topography was not very high (Karner and Gambôa, 2007). Precipitation rates may indeed have varied with time of deposition. Fast rates are associated to higher K_d , and depending on the age model we use the Alagoas carbonates had either lower or greater rates of deposition than those from the Jiquiá, as discussed above. Thus, it is not straightforward to assess the role of the rates of precipitation as a relevant controlling factor. On the other hand, when the Sr/Ca ratio increases in the liquid phase, Sr concentration tends to increase proportionally in the solid (Jacobson and Uzdowski, 1976). This supposition seems more likely in the geological context that is being considered and apparently agrees with observations.

Additionally, the $^{87}\text{Sr}/^{86}\text{Sr}$ vs. Mn/Sr data in Fig. 6C suggests two trends: one, in the lower part of the diagram, which may represent the depositional history, in agreement with the Rb/Sr and $1/[\text{Sr}]$ data, and another, in the upper part, that seems to be indicative of the interaction with post-depositional fluids modifying the primary carbonates. The latter trend suggests that the system may have become open and prone to dissolution and recrystallization of primary carbonate minerals, incorporating Mn ions, likely in reducing conditions, where the smaller ionic radius of Mn^{2+} would not make it unfavourable to assimilation into the carbonate lattice. It is also seen from $^{87}\text{Sr}/^{86}\text{Sr}$ vs. Mg/Ca in Fig. 6E, that Mg/Ca ratio is usually rather low for the majority of samples, seemingly representing a low background value of < 1 . This allows discriminating a set of a few samples that were likely subjected to pervasive recrystallization (within the ellipse area), without apparently, at least significantly, being able to alter the original $^{87}\text{Sr}/^{86}\text{Sr}$ isotopic composition. Alternatively, dolomite could have formed very early, at later stages during the deposition or during early diagenesis. Either way, such a few samples presenting signs of alteration, and the consistency of the $^{87}\text{Sr}/^{86}\text{Sr}$ values suggest that post-depositional fluids could not significantly overprint the primary isotopic signals, and that the system was likely rock-buffered with respect to any later fluids. Although the Mg/Ca ratio represents bulk rock analysis, what is clear from the background values is that most of the carbonate phase present in the samples is consisted of calcite, and dolomite constitutes the subordinate carbonate mineral phase, as also observed from petrography. From Fig. 6F it is observed that dolomite samples do not contain expressive concentrations of Mn, and the opposite seems true for some of the samples composed by calcite. With respect to the three samples with significantly high Mg/Ca ratio, another possibility is that they may constitute a mixture of carbonate and somewhat labile Mg-clay minerals, because the latter may have been prone to dissolution by the digestion method that was used (i.e., aqua regia dissolution). This may be clarified in future X-ray diffraction studies.

It is not possible to securely infer, based on bulk chemistry alone, if the primary precursor carbonate phases were aragonite or calcite, or stable calcite or Mg-calcite. The considerable proportion of dolomite observed in some samples and the occurrence of Mg-silicates, as illustrated in thin-sections shown in Fig. 3, apparently suggest that the original fluid Mg/Ca ratio was likely relatively high. In such circumstances, the precipitation of more pure Ca-carbonate, in the form of aragonite, is favoured, leading to even higher fluid Mg/Ca ratios (Deocampo, 2010). In this case, dolomite would not represent a primary constituent in part because of the larger hydration sphere of Mg compared to Ca, resulting in kinetically unfavourable conditions to

dehydrate as required for fixation on mineral surfaces (Reeder, 1983; Deocampo, 2010). As a result, dolomite may have formed in very early diagenetic stage, in periods of desiccation or fluid expulsion amidst the sediment, due to compaction and/or crystal growth of later phases. Nonetheless, in terms of its Sr isotopic composition, it should be reasonable to expect that later phases such as dolomites would not differ significantly from primary carbonate phases, because they would derive from the same but more concentrated fluid.

Notwithstanding, apart from the bivalves, whose shell composition vary and can be composed solely of aragonite (Scholle and Ulmer-Scholle, 2003), the original constitution of the main carbonate content now comprised by calcite minerals is mostly speculative. Although the chemical analyses reveal Sr concentrations in the pre-salt carbonate samples compatible with those in modern calcite rather than aragonite minerals precipitated from the same waters (e.g., Gasse and Fontes, 1989; Sunagawa et al., 2007), the possibility of loss of Sr due to inversion of aragonite to a stable phase could not be entirely excluded. However, facies constituted by calcite shrubs (Fig. 3) do not show evidence of previous microstructures and crystal dissolution and substitution, which would be required for replacement of aragonite by calcite (Morse and Mackenzie, 1990). In addition, the significant amount of bulk rock data and its consistency throughout the geological sequence allow us to have confidence with respect to its significance as a reliable proxy for general lake water chemistry and hydrology. Apparently, roughly 10 samples are suspected to have been subjected to more significant alteration according to the data displayed in Fig. 6, and these are also highlighted in the supplementary material table (available upon request). Despite this, they have not been excluded from the whole set of data for modelling purposes. In general, along with petrographic observations, which allow distinguishing the presence of calcite from dolomite, it seems that the possibility of pervasive recrystallization significantly modifying the original rock isotopic composition can be disregarded. Hence, the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio can be used as a proxy for palaeohydrologic reconstruction of the lake through time.

4.4. Strontium isotope data

To account for radiogenic contribution derived from detrital components we corrected our measured $^{87}\text{Sr}/^{86}\text{Sr}$ (R_{meas}) to the initial $^{87}\text{Sr}/^{86}\text{Sr}$ (R_{initial}) using calculated $^{87}\text{Rb}/^{86}\text{Sr}$ values based on the ratio $^{86}\text{Sr}/^{88}\text{Sr} = 0.1194$ and the abundances of the other naturally occurring Sr isotopes in our samples (Faure and Mensing, 2005). The Rb/Sr results provided by the chemical analyses, discussed above, also necessarily had to be accounted for in this correction. The correction took into account the ^{87}Rb decay constant of $1.42 \times 10^{-11} \text{ y}^{-1}$ and a perhaps overestimated age of 130 Ma. We chose to consider this age due to uncertainty regarding the absence of sufficient precise age constraints with respect to the pre-salt deposits and to deliberately extrapolate the maximum age of our section to an older limit. If no significant change results from this extrapolation, then it means modification of the original lake water $^{87}\text{Sr}/^{86}\text{Sr}$ and thus of the carbonates due to detrital contribution is of little concern and can be disregarded. The curve based on the corrected values is also shown in Fig. 7. It can be seen that the correction does not result in significant change for most of the profile, except for the bottom interval, where some of the corrected data become considerably less radiogenic.

Although Richter and DePaolo (1988) and Richter and Liang (1993) have comprehensively addressed the problem of diagenesis affecting the original $^{87}\text{Sr}/^{86}\text{Sr}$ composition of marine carbonates, they concluded that this effect is small for conditions where high-precision

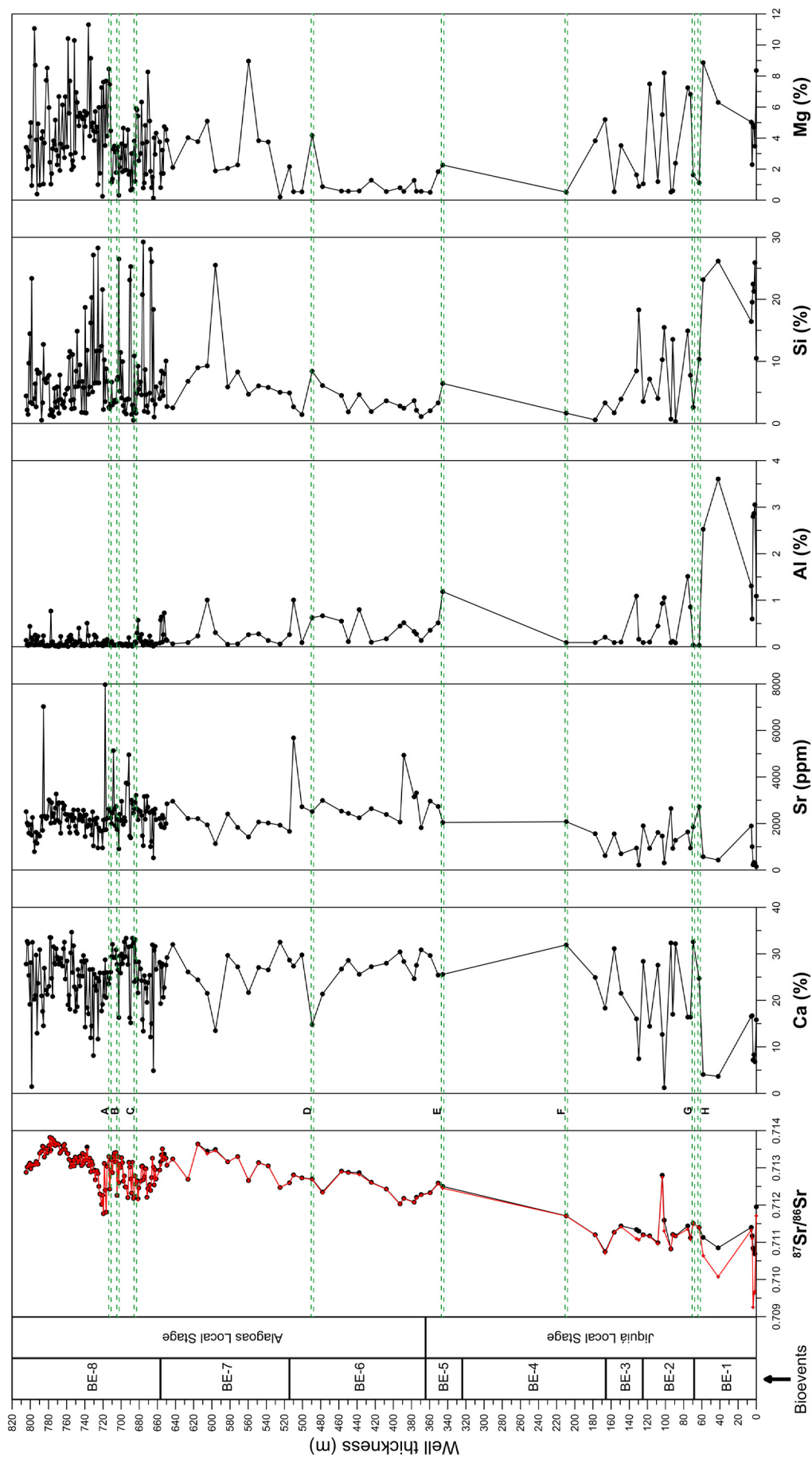


Fig. 7. Measured $^{87}\text{Sr}/^{86}\text{Sr}$ composition in the 194 carbonate-rich rock samples (black) and corrected ratio for initial $^{87}\text{Sr}/^{86}\text{Sr}$ composition (red) with respect to well thickness, along with the concentration profiles of Ca, Sr, Al, Si and Mg. The sample at the bottom of the well was arbitrarily placed at zero metre depth. The bioevents and the local stages based on non-marine ostracods are indicated to the left. The bioevents were established based on specimens recovered from cuttings samples. Letters A to H refer to samples from which photomicrographs shown in Fig. 4 were taken. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

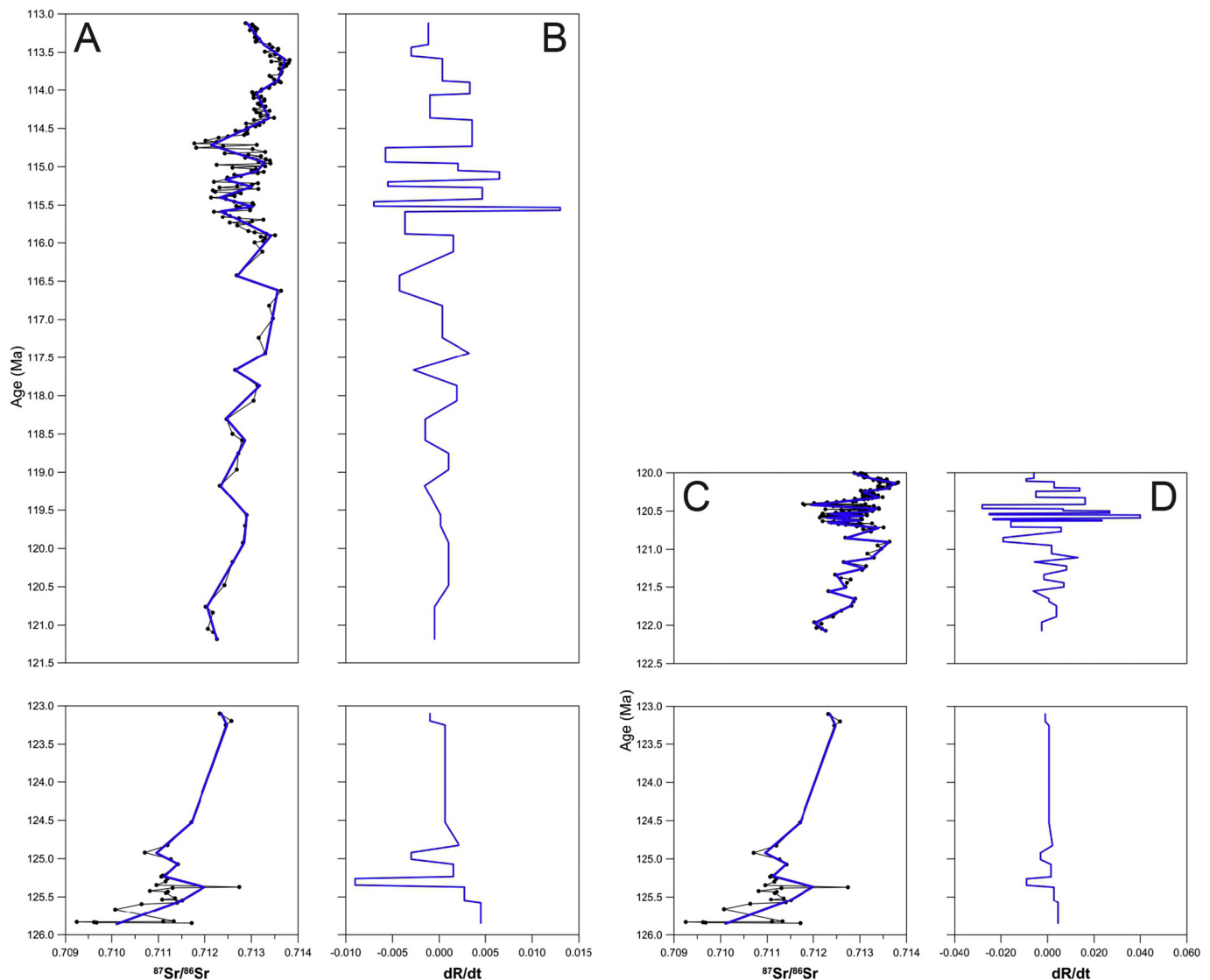


Fig. 8. Corrected ratios for initial $^{87}\text{Sr}/^{86}\text{Sr}$ composition of the 194 carbonate-rich rock samples relative to age estimations based on different literature sources. A: Ages according to Moreira et al. (2007), Winter et al. (2007) and biozone calibration to international chronostratigraphy after Poropat and Colin (2012b) and references therein. C: Age estimation after Tedeschi et al. (2017). The smoothed curves of carbonate $^{87}\text{Sr}/^{86}\text{Sr}$ are based on the corrected ratio for initial $^{87}\text{Sr}/^{86}\text{Sr}$ ratio composition, and are shown as thick blue lines. B and D: Rates of variation of the $^{87}\text{Sr}/^{86}\text{Sr}$ of lake water through time (first derivative of the respective simplified smoothed $^{87}\text{Sr}/^{86}\text{Sr}$ curves). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

stratigraphic correlation is not the main target of the study or it is not required at all. In the present case, the carbonates were deposited in continental environment. Although the salt immediately above may be entirely marine in origin, we do not address this question in our study since no samples of the salt layer were available and it is beyond the scope of this work. Neither were samples of formation water available to evaluate for buffering capacity between pore water and sediment, and verify if they are in equilibrium at any given point. In a drilling program, formation water is hardly a target worth of sampling due to time constraints. In addition, from the geological point of view, a lake, irrespective of its size, is a much more dynamic system than the oceans, with residence times of the elements significantly shorter than those found in the latter. For both the above reasons and those discussed in the previous item, we refrain from using our isotopic data to the very last digit of analytical precision and instead take into consideration up to the fourth significant figure of the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio. Very high frequency variations were also disregarded, and thus we adopted a smoothed curve for the purposes of modelling in both age model scenarios 1 and 2 (Fig. 8A/C). Notwithstanding, another observation in

favour of the preservation of the Sr isotope ratio, for example, comes from some of the samples of the Alagoas sequence. As described earlier and illustrated in Fig. 4, the calcite crystal-shrubs limestones are common to this sequence, and samples composed by finer fabrics and/or containing more than a single carbonate mineral phase, such as dolomite, do not differ expressively from adjacent samples essentially represented by the former.

From Fig. 8 it was possible to obtain the rates of change of the $^{87}\text{Sr}/^{86}\text{Sr}$ with respect to time for scenarios 1 and 2, shown in Fig. 8B/D, respectively. One can observe that age model scenario 2 results in rates of change roughly one order of magnitude greater than age model scenario 1.

Apparently, we can also disregard a significant contribution from diagenesis of the carbonate rocks below (i.e., the coquinas) according to the geochemical data, discussed previously, and assuming that those rocks have been lithified very early. The existence of the pre-Alagoas unconformity between the two main rock units implies that a significant amount of time elapsed. Thus, while influx of interstitial waters from the carbonates of the sequence immediately below may have been

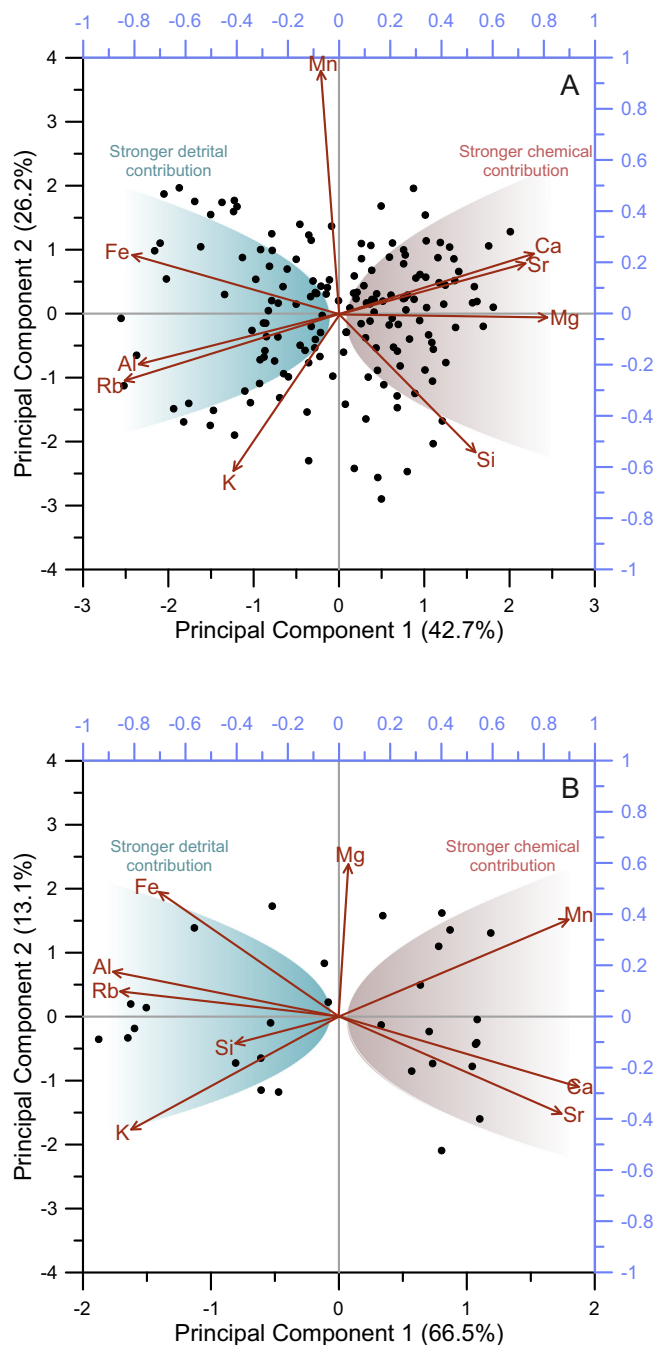


Fig. 9. Results from the principal component analysis. A: Loadings and scores for samples from the Alagoas. B: The same for samples from the Jiquiá. The fields to the left and right indicate components more directly related to detrital contribution (i.e., the “mechanical” particles) and to chemical contribution (cations precipitated from solution), respectively.

possible, that part of the system was likely kept buffered or did not contributed significantly as a source of ions due to its subsidence and compartmentalisation within the basin.

4.5. PCA results as supporting data for the model

A principal component analysis was performed with the entire geochemical data set obtained previously. The results are presented in Fig. 9. The lower plot represents the results for Jiquiá samples, and the upper plot corresponds to results of the PCA for samples from the Alagoas. The most interesting aspect of these plots is that they both

clearly divide two very distinct fields, one comprising samples more influenced by terrigenous siliciclastic contribution, to the left, and another with samples more strictly related to chemical deposition, to the right. With respect to the Jiquiá interval, it can be observed that its samples present greater silicate content likely associated with constituents of detrital nature, also shown by the chemical analysis and the PCA results. As shown above, the environment in general seems to have been relatively milder from the point of view of micropalaeontology, which reveals a greater abundance of non-marine ostracods in the lower part of the studied section, followed by a steady trend of decreasing occurrences and forms upwards. Together, these data support the assumption that fluvial input should have been more significant during that time.

Regarding the upper interval pertaining to the Alagoas local Stage, the carbonates correspond essentially to authigenic precipitates, seldom containing mud-sized detrital siliciclastic material and rare bioclasts, represented by ostracod shells. This can be deduced also from the geochemical standpoint, as expressed in the PCA results. Whereas the general distribution of elements in both graphics A and B from Fig. 9 shows a similar pattern, there are subtle differences between them. Rubidium, aluminium, iron and potassium are in both cases related to the field that corresponds to stronger detrital contribution, on the left corner of the graphics. Silicon, on the other hand, is closely associated to those four elements in samples belonging to the Jiquiá, whereas it shows an adverse behaviour when we consider the samples that belong to the Alagoas. Such a difference can be attributed to the greater occurrence of chert beds in the Alagoas (Fig. 9A), with silica cementing pores or replacing previous mineral phases. Some minor terrigenous constituents are also present in the samples from this interval. This could be the reason why silicon did not correlate with principal component 1 (PC1) to great extent, because it is not predominantly linked to detrital constituents. In addition, its soluble forms require distinct physical-chemical conditions from those of carbonates to precipitate. Probably for a similar reason, manganese do not correlate with any other element in the Alagoas, because it might be predominantly associated to phases other than carbonates, such as oxides. In the Jiquiá (Fig. 9B) Mn shows greater correlation with Ca and Sr, suggesting that these carbonates might have formed incorporating some of this element in their structure, a feature that could be further investigated using high-resolution chemical analyses associated to cathodoluminescence in petrographic thin-sections. A similar pattern was already indicated in Fig. 6, which highlighted $^{87}\text{Sr}/^{86}\text{Sr}$ vs. Mn/Sr ratio and showed signs of two distinct Mn contributions.

The PCA results clearly show that this methodology can be applied preliminarily as a means of exploratory data assessment and to indicate ways to conduct further specific analysis in a given study. Accordingly, the geochemical data and the PCA results corroborate petrographic observations and the expected general behaviour of the elements. We can expect, additionally, carbonate samples from Jiquiá to be relatively more calcitic, due to greater correlation of Ca and Sr alone with PC1, and carbonates from the Alagoas to contain a greater proportion of dolomite, as shown by the association among Ca, Sr and Mg in Fig. 9A. This final observation is also supported by the examination of thin-sections.

4.6. Box-model set-up

Based on biostratigraphic evidence above, we know that the lowermost part of the studied interval presented greater faunal abundance and diversity, with characteristics of species living in a more suitable, less stressing environment, which suggests a more humid climate. Upward in the section and in the stratigraphy these characteristics change significantly, and the fauna becomes less abundant and diverse and the remaining species show thinner skinned shells, typical of a species living in more severe lake water conditions, possibly associated with a change in the local climate and/or tectonic environment.

Along with the characteristic facies variation upwards and the geochemical data discussed previously, it can be deduced that whatever the reason for the significant change observed in the biostratigraphic record and in sedimentation pattern was, these changes reflect an alteration in the hydrology of the lake. These observations indicate that the lake system was likely fed by a river or drainage system, which contributed relatively more diluted waters to the lake and allowed a more diverse ostracod fauna to thrive. Its relative influence possibly became progressively less effective during the evolution to the sag phase, as a resulting lower relief would affect negatively the flow rate of surface waters and the source catchment area probably became further far away. This interpretation is compatible with the evolution of the tectonic environment, in which the lake would initially be embedded in a narrower rift valley and evolve to a lake enclosed within a progressively wider basin during the sag phase.

To test this scenario in a numerical model a few other additional constraints must be addressed first. The numerical values chosen to run the mass-balance models based on Eq. (4) are bound to a limited array of data acquired directly from the sampled material and to a few assumptions that need to be made, and which are detailed below.

First, there is a lack of direct evidence with respect to the precise lithological composition of the basement of Santos Basin. Information regarding the isotopic composition of these rocks is even scarcer. To the best of our knowledge, the best available data come from published studies referring to outcrops along the coast of Rio de Janeiro State and from inferences based upon aeromagnetic data. For instance, studies indicate that orthogneisses of tonalitic to syenogranitic composition associated to high-grade metamorphic rocks constitute the Precambrian basement of Santos Basin, belonging to the Cabo Frio tectonic domain (Schmitt et al., 2008; Zalán and Oliveira, 2005). The origin of this domain is not well known, and it has been proposed that it would be related to the São Francisco Craton margin, a relict of the Congo Craton in West Africa or to a separate microcontinent (Valladares et al., 2008; Schmitt et al., 2008). The most incisive evidence regarding the nature and the Sr isotopic composition of the basement rocks comes from the study of Tupinambá et al. (2012), who studied in greater detail the rocks of the Rio Negro Complex of the Ribeira Fold Belt. The latter consists of a plutonic portion of a magmatic arc emplaced by the subduction of the ESE border of the São Francisco palaeoplate during the amalgamation of the Western Gondwana. These authors have shown a variable range of $^{87}\text{Sr}/^{86}\text{Sr}$ ratio values, covering the spectrum from more mafic components to more felsic ones (Tupinambá et al., 2012). Tedeschi (2017) recalculated the $^{87}\text{Sr}/^{86}\text{Sr}$ ratios provided by the latter authors to ages of 113 and 130 Ma, showing this suite of Neoproterozoic rocks should have had $^{87}\text{Sr}/^{86}\text{Sr}$ ratios in the range between approximately 0.704–0.735, with a median value of 0.711 at 113 Ma and a slightly higher median of ca. 0.713 at 130 Ma.

There is also evidence for the flood basalts that overlie Precambrian basement in the Campos Basin and the nearby flood basalts from the Paraná Basin. Both have been sampled and comprise tholeiitic basalts, with corrected $^{87}\text{Sr}/^{86}\text{Sr}$ from the former ranging between 0.70586 and 0.70926 and a single sample with 0.71892 (Mizusaki et al., 1992), whereas the latter shows a larger range in isotopic composition, from 0.70483 to 0.71151 (Peate and Hawkesworth, 1996; Peate et al., 1999). The basalts of the Camboriú Formation in the Santos Basin are stratigraphically correlated to those of the Campos Basin (Moreira et al., 2007), in such a way that similar isotopic compositions may be expected. More recently, there has been the suggestion, based on seismic interpretation, that the basalts of the Paraná Basin, as well as a significant sedimentary succession of the latter, may constitute part of the basement of the Santos Basin (Zalán, 2017).

Another evidence concerning the palaeo-lake basement rocks comes from recent studies published by Wright (2012) and Wright and Barnett (2015) that discuss the occurrence of Mg-silicate minerals in association with the carbonates. The authors point out that the occurrence of these tri-octahedral silicates together with calcite is an indication of lakes

sourced by volcanic terranes. But as we see from the above $^{87}\text{Sr}/^{86}\text{Sr}$ data determined in our carbonate samples, volcanic sources probably were not the only source to the lake. Felsic rocks from the basement and/or alluvial material derived from it were likely also contributing as sources of ions to the composition of subsurface waters, as suggested by Tedeschi (2017), or perhaps as another catchment area draining to the lake. However, concerning this latter argument the lack of detrital components in expressive amounts in the corresponding sequence does not support the hypothesis of direct input resulting from erosion and surface transport via catchments and rivers.

On the other hand, the relative contribution made from waters draining or seeping through felsic rocks is speculative and can only be inferred. We do not know the exact configuration of the palaeodrainage basin, or of the groundwater system. However, we can assume some reasonable premises. For instance, Reston (2010) argues that the upper surface of a lake in a rift setting would initially be maintained above sea level as a result of initial thermal uplift and high volume of sediment influx. Consequently, excess water would drain from the rift-valley edges towards the open ocean. With further regional subsidence the basin would become internally draining and the lake surface would stand below sea level. Following this, the lake water would become more brackish and evaporation would become the only way to remove water from the basin. The thought is reasonable and is upon this scenario that we build our model.

Geophysical investigation based on gravimetric, magnetic and seismic data supports the postulate that differentiated mantle-derived magma underplating added material to the highly stretched continental crust on the Brazilian southeastern margin below the structural high named the Outer High (Carminatti et al., 2008; Gomes et al., 2009; Scotchman et al., 2010; Buckley et al., 2015). Accordingly, this resulted in long lasting positive structures on which it is assumed that the deposition of shallow-water lacustrine carbonates was favoured, recording an almost continuous deposition from Barremian to the Aptian, such as the Tupi and Sugar Loaf highs (Mohriak et al., 2012; Buckley et al., 2015).

Additionally, lakes in active rift settings are occasionally fed by subsurface waters of both hydrothermal origin or associated with groundwater seepages. Hydrothermal systems are known to occur in modern-day lakes in rift settings (Tiercelin et al., 1989; Barrat et al., 2000; Dekov et al., 2014), and there is compelling evidence in Campos Basin suggesting that the same might have occurred in Santos Basin (Alvarenga et al., 2016). Souza et al. (2018) have also given support for the hydrothermal origin of the Barra Velha carbonates in the Santos Basin. Besides this, the supply of subsurface waters, whether hydrothermal in origin or not, can be significant, as the existence of several lakes in arid zones is a testimony (Hillaire-Marcel and Casanova, 1987; Gasse and Fontes, 1989; Casanova and Hillaire-Marcel, 1992a, 1992b; Kebede et al., 2005; Kebede et al., 2008). Moreover, Tedeschi (2017) has shown more direct evidence of significantly radiogenic groundwaters, extensively influencing the isotope values of the lake-derived carbonates spanning deposits from the basins of Campos to Santos.

According to the evidence provided by his study and to additional observations, direct surface seawater incursion into the pre-salt lake can be discarded, at least significantly, because: (1) strontium isotopes indicate the opposite trend than that one would expect for an opening seaway; (2) in the studied well there are no occurrences of sulphate or chloride deposits interbedded with the carbonates along this interval, both as primary or diagenetic precipitates, not even as pseudomorphs or as engulfed remnants after complete or partial replacement by other diagenetic phases, or moldic porosity after complete dissolution of salt crystals; (3) the absence of unequivocal marine fossils; and (4) according to published data during most of this time the Florianópolis and Rio Grande Highs to the south of the basin appear to have prevented the access of marine waters, which finally seems to have occurred from the north during the late Aptian (Arai, 2009; Scotchman et al., 2010; Thompson et al., 2015; Buckley et al., 2015). The synthesis of the

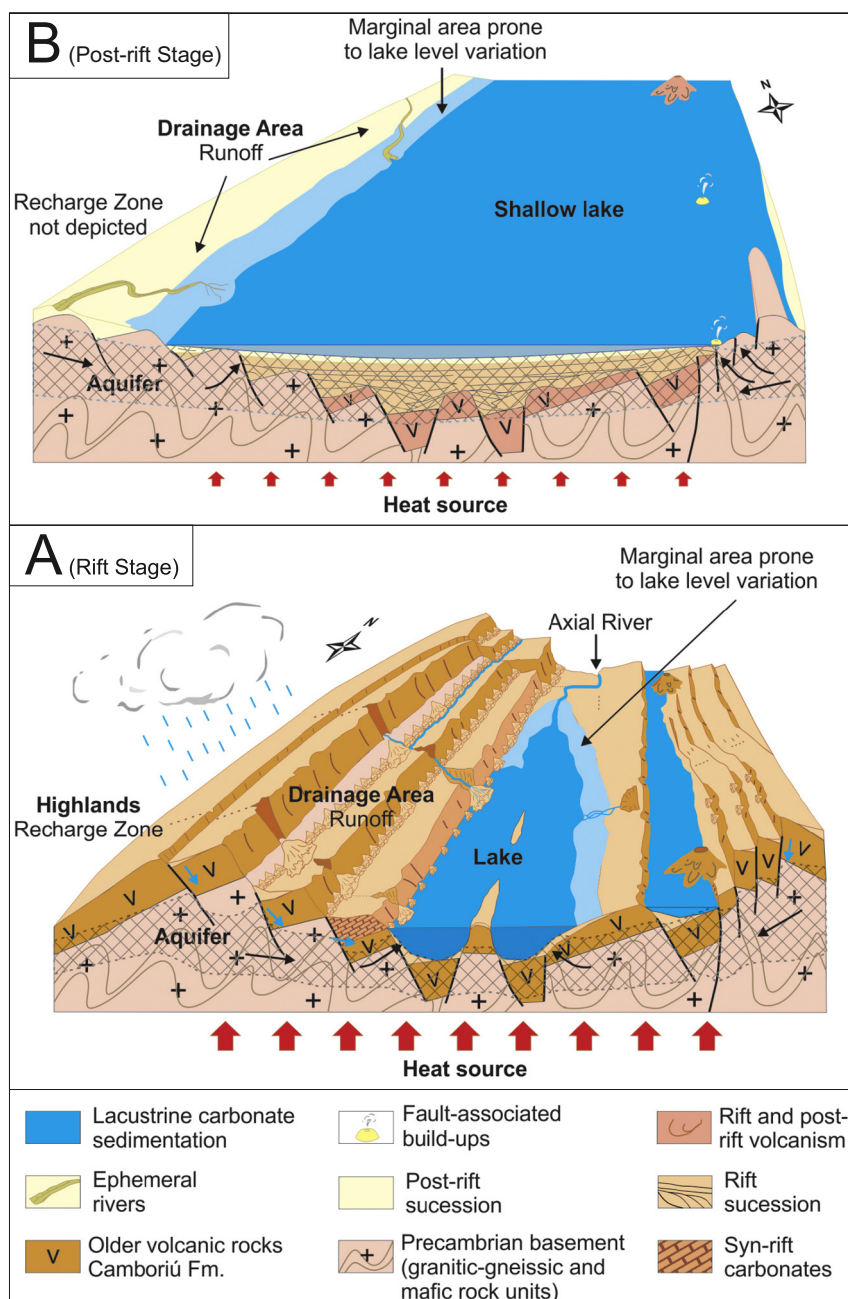


Fig. 10. Simplified conceptual model illustrating the general geology and hydrology of the pre-salt rift system. Small blue arrows show infiltration to and recharge of the aquifers; long black arrows indicate infiltration of groundwater to the lake basin. Thick red arrows underneath represent geothermal heat, which is assumed to decrease from rift to post-rift phase. Hachured areas propose the presence of aquifers relative to the lake basin, flowing mainly through fractured crystalline basement. Initial conditions for calculation in the numerical models are: N (lake water $[Sr]) = 39 \mu\text{mol/L}$; R_f (average felsic basement $^{87}\text{Sr}/^{86}\text{Sr}$ ratio) = 0.714; R_m (average mafic basement/latter volcanic rocks $^{87}\text{Sr}/^{86}\text{Sr}$ ratio) = 0.706; $F_f = 7.39 \times 10^6 \text{ mol Sr/year}$; $F_m = 29.45 \times 10^6 \text{ mol Sr/year}$. A) Idealization of the rift phase environment, corresponding to the time of deposition of the Jiquiá sequence, where a deeper lake formed, which could eventually be subjected to water-column stratification. B) Idealization of the post-rift phase environment, at the time of deposition of the Alagoas (Barra Velha) sequence, where a shallower, endorheic lake with a large areal coverage evolved. Illustrations are not to scale. The initial fluxes F_f and F_m given above are used in the calculations whose results are displayed in Fig. 11, lower axes. These parameters are reversed (i.e., $F_f = 29.45 \times 10^6 \text{ mol Sr/year}$ and $F_m = 7.39 \times 10^6 \text{ mol Sr/year}$) for calculations whose results are shown in Fig. 11, upper axes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

geological observations described above is represented in the conceptual geological model illustrated in Fig. 10.

It is rather difficult to estimate the fluxes from catchment areas and/or hydrothermal vents that may have contributed to the lake basin. One likely driver of hydrothermal influx, crustal heat flow, probably changed over time in the area during the time span considered as thermal subsidence progressed (Karner and Gambôa, 2007; Reston, 2010). As a result, hydrothermal flux probably varied as a function of this change. Additionally, influx of groundwater other than hydrothermal water is essentially function of hydraulic head, rather than heat flow. Therefore, if either hydraulic head or heat flow or both decreased with time, which is possible, because rift shoulders are eroded and crustal cooling occurs, it is reasonable to assume that the input parameters F_f and F_m in Eq. (4) should respond in accordance to that assumptions, i.e., both should decrease with time. We then consider three scenarios in this respect, one that takes F_f as constant through time, a second in which this parameter constantly decreases with time until it

reaches half its initial value and another in which it decreases one order of magnitude from the initial value assumed. We do the same with respect to F_m to calculate the model dependent on this term. Differently from other published approaches (e.g., Doebbert et al., 2014), the effects of variable evaporation and precipitation fluxes directly impacting the lake volume are subjectively implied in this scenario.

Hence, our choice was to consider a present-day lake from similar latitudinal range and possibly also similar area and initial lake volume, in an equivalent initial rift scenario. The Lake Tanganyika seems suitable for this purpose, as already illustrated in Fig. 1 above. For modelling purposes, strontium concentration in lake waters was taken from reference lakes in Africa, the Tanganyika itself and the Awash Basin in Ethiopia (Barrat et al., 2000; Bretzler et al., 2011), assuming an average $N = 39 \mu\text{mol/L}$. It could be argued that this choice could possibly not be representative, since bedrock and surrounding lithologies might differ between present-day African locations and that of the Brazilian pre-salt lake, which would naturally result in distinct contributions of Sr.

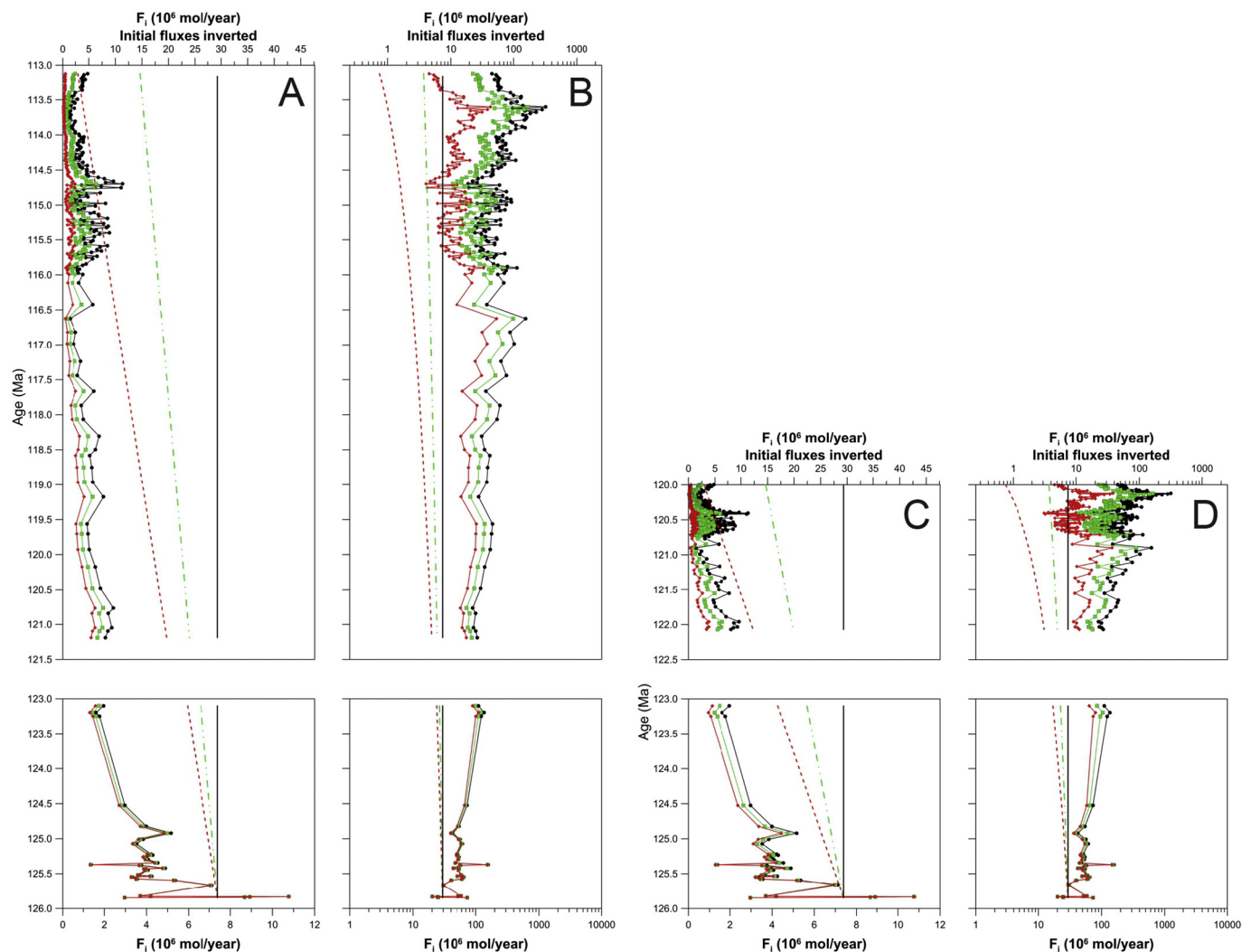


Fig. 11. Lower axes: A) Fluxes of [Sr] due to mafic sources when the contribution from felsic sources is held constant through the time span covered (black circles and black dotted line, respectively), and when the latter decreases half its initial value and one order of magnitude from initial value at constant rate (green squares with dotted and dashed line and red diamonds with red dashed line, respectively), according to the model parameters explained in the text; B) Fluxes of [Sr] due to felsic sources when the influx derived from mafic sources is held constant through the time span covered (black circles and black dotted line, respectively) and when the latter decreases half its initial value and one order of magnitude from initial value at constant rate (green squares with dotted and dashed line and red diamonds with red dashed line, respectively), according to the model parameters explained in the text; C) The same as in A, but considering the age estimation for scenario 2; D) The same as in B, but considering the age estimation for scenario 2. Upper axes: A, B, C and D are the same as above, except for the initial values of F_f and F_m , which were swapped. Thus, here $F_f = 29.45 \times 10^6$ mol Sr/year and $F_m = 7.39 \times 10^6$ mol Sr/year are the initial values set to run the models. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Notwithstanding, changing the value of N in Eq. (4) would only produce a significant change in the results for very large values of dR/dt (Richter et al., 1992). Regardless of that, it can be easily verified that changing N a few orders of magnitude to higher or lower values does not change the final result significantly. To account for the influx derived from felsic sources and Sr concentration in these fluids, we assumed as the initial value the same as that of hydrothermal flow reported by Tiercelin et al. (1989) and hydrothermal fluid concentration of Sr determined by Barrat et al. (2000) for the Tanganyika (3 L/s and 78.1 mmol/kg, respectively). This gives for the initial flux derived from felsic sources (F_f) approximately 7.4×10^6 mol/year. In order to estimate the overall influx derived from mafic sources, F_m , from input flux from felsic sources F_f , three conditions are considered: one in which F_f is kept constant throughout the entire studied interval, a second in which it decreases constantly with time until it reaches half its initial value, and a third in which it decreases one order of magnitude from beginning to end, as indicated before.

Since we are considering an environment where riverine influx was

not very high from the start, as indicated from the absence of fossil fauna other than bivalves and ostracods, as discussed above, we introduced into our model an initial flux compatible with that of a river draining through an arid landscape, such as the Blue Nile. One must bear in mind that this is for practical purposes being considered conceptually equivalent to the flux derived from mafic source rocks, which might as well have comprised a source for subsurface waters flowing to the lake. Conversely, the same idea should apply to the felsic sources, which might have been exposed to weathering under surface conditions. These ideas are further detailed below, whose implications are discussed more deeply with aid of the numerical models.

Regardless of these uncertainties, we need a starting point and we assumed for the influx derived from mafic sources F_m , a flux equal to 29.45×10^6 mol/year (Palmer and Edmond, 1989). Similarly to what was described above for the influx derived from felsic sources, we also use three scenarios in the simulation with fixed F_m as input parameter: one in which it is constant through time, a second in which it decreases constantly with time until reaching half its initial value, and another in

which it decreases one order of magnitude from beginning to end.

The lake water $^{87}\text{Sr}/^{86}\text{Sr}$ ratio (R_{lw}) is taken as equal to the corrected initial $^{87}\text{Sr}/^{86}\text{Sr}$ ratio determined from the analyses of the carbonates samples. Because the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio is not subjected to isotopic fractionation, the depositional signal is not expected to have changed significantly. Unless a later fluid with distinct isotopic ratio interacted with the host rocks, represented by the deposited material, favouring dissolution and recrystallization of later phases with a different isotopic composition. However, as extensively discussed above, the geochemical data and petrographic observations, along with the recovery of well-preserved fossils, are conspicuous evidences that support the view that no significant alteration occurred under burial regime and therefore these rocks generally preserve their depositional Sr isotopic composition.

In order to reproduce the corrected strontium isotope curve and to avoid meaningless output (i.e., negative fluxes) we are not allowed many degrees of freedom with respect to the choice of the input parameters if we strictly consider two end members alone. Following this reasoning, the choice of R_m and R_f cannot be entirely arbitrary even if there are not enough data to constrain an average value for the overall lithologies that constituted the basin in which the lake was enclosed. Notwithstanding, the values we chose are not unrealistic and their choice is considerably conservative, with $R_m = 0.706$ and $R_f = 0.714$, close to the median values of the end-members established so far. The average value of R_m could have been lower as seen from available $^{87}\text{Sr}/^{86}\text{Sr}$ data from basalts and other mafic rocks in adjacent areas (Mizusaki et al., 1992; Peate and Hawkesworth, 1996; Peate et al., 1999; Tupinambá et al., 2012). Moreover, the average value of R_f could have been considerably higher than the one adopted here, as observed from data published by Tupinambá et al. (2012). However, the choice of values within a close range to those ones does not result in any significant modification of the output trends, the latter being our main concern, rather than inferring as precise as possible absolute flux values.

The results from these distinct conditions are plotted in Fig. 11A/B (age model scenario 1) and Fig. 11C/D (age model scenario 2), respectively. We can see that, for the established initial input of Sr equal to 7.4×10^6 mol/year for F_f , there is a peak influx F_m at the bottom of the interval in the order of 11×10^6 mol/year (Fig. 11A/C). The continuous, dotted and dashed, and dashed lines represent the input F_f values, and the curves with black circles, green squares, and red diamonds the fluxes derived from mafic sources that are needed to account in the given conditions for the $^{87}\text{Sr}/^{86}\text{Sr}$ that we have determined in the carbonate samples. It can be observed that, in general, F_m decreases with time in all circumstances, although a few lower pulses occur in the Jiquiá and in the Alagoas. In the case in which the flux F_f is left to decrease constantly one order of magnitude with time (red dashed line), F_m reaches a minimum modelled range in the order of $0.3\text{--}1 \times 10^5$ mol/year close to the top of the studied succession, remaining nearly within this range until the salt above. Considering the three different modelled conditions, the overall modelled range of the flux F_m lies between 0.2×10^5 and 11×10^6 mol/year (with median $0.2\text{--}1 \times 10^6$ mol/year) during the span of time considered.

In Fig. 11B/D the input parameters are reversed. Fluxes derived from mafic sources are either held constant or allowed to decrease half its initial value and one order of magnitude with time from the initial value of 29.45×10^6 mol/year (continuous, dotted and dashed, and dashed lines, respectively). The corresponding fluxes derived from felsic sources are represented by the curves with black circles, green squares, and red diamonds, respectively. Overall, the latter fluxes vary within the range of ca. $16\text{--}1274 \times 10^6$ mol/year (median $52\text{--}203 \times 10^6$ mol/year between models). In the models in which F_f is allowed to decrease at regular rates, from the starting condition of 7.4×10^6 mol/year, the F_m also decreases fairly regularly, presenting some pulses along the way (Fig. 11A/C). Nevertheless, both fluxes conform in the sense that they continuously decrease with time, in the

range below 10×10^6 mol/year. Conversely, in the case in which F_m has the starting condition of 29.45×10^6 mol/year and decreases regularly to the top (but no less than one order of magnitude), F_f actually increases considerably in some intervals in order to maintain the values of $^{87}\text{Sr}/^{86}\text{Sr}$ observed in the carbonate samples (Fig. 11B/D). Regardless of this difference in particular, both sets of modelled conditions show that F_f should have become the prevailing flux very early in the history of evolution of the basin, as determined from these initial conditions.

This difference between the two main results (A/C and B/D) also highlights that at least two distinct conditions might explain the observed $^{87}\text{Sr}/^{86}\text{Sr}$ values in the carbonates: the decrease of both F_f and F_m with time suggests that, depending on the lithology exposed or underlying the lake basin, either runoff or groundwater influx, or both, decreased (results A and C, in which F_m shows an overall decrease for all the three conditions considered for F_f). On the other hand, results B and D show in general an increase in the values of F_f with time, at the expenses of progressively lower F_m values. The suggestion in the models derived from the initial conditions in Fig. 11B/D is that the flux F_f was predominantly derived from *groundwaters* in contact with felsic rocks, since there is no sedimentologic evidence for detrital transport. From approximately the middle part of the Barra Velha Fm. onwards F_f falls abruptly, but recovers towards the top, although there are recurrent falls along the way. Several episodes of greater fluxes appear to have occurred, reaching its greater value of 1274×10^6 mol/year (in the model considering constant fluxes derived from mafic sources – black circles in Fig. 11B/D), after which F_f decreases abruptly towards the top of the studied sequence. The flux of Sr from felsic sources apparently decreased to a “background” value, no greater than ca. 15 and 190×10^6 mol/year in the uppermost Barra Velha Fm., considering the corresponding flux of F_m that yields the bigger F_f values (i.e., constant F_m through time). Considering the other extreme used in these models, i.e., F_m decreasing one order of magnitude, F_f falls more pronouncedly after the intermediate portion of the Barra Velha Fm., despite showing the same high frequency pattern between alternating episodes of fall and followed by increase in its fluxes (red diamonds in Fig. 11B/D).

Notwithstanding, the compilation of all results shows that F_f should have been indeed higher than F_m for most of the time, and might even have increased relative to the latter during the evolution of the lake system. From the point of view of the data discussed so far and the geological evolution of the basin, this scenario implies that whatever the origin of the fluids was, meteoric or groundwater, they likely reacted progressively more with felsic rocks with evolution of time. Since other geological observations suggest low riverine input, as derived from sedimentological and micropalaeontological data, for instance, this might be an indication that water recharge to the lake occurred mainly from groundwater interacting with basement rocks or their detrital derivatives. Hence, the synthesis provided by the quantitative models, that better adjusts to the geochemical, sedimentological and micropalaeontological observations, points to progressively lower riverine and somewhat subtle declining subsurface fluxes with time, in a broader sense. Eventually, in shorter-term cycles, fluxes may have increased momentarily, although the main trend seems to have been of general decrease.

One point worth mentioning is that testing the models for the two scenarios with respect to time and all the rest being equal (A versus C and B versus D) shows that the influx trends are conserved, generally resulting only in negligible changes in the corresponding magnitudes. Although the trends in both scenarios are equivalent, the rate of change in scenario 2 is nearly one order of magnitude greater in general. Hence this is a necessary consequence that the influxes of [Sr] would have to change significantly faster in the latter case in order to maintain the isotopic mass-balance in the lake.

In yet another calculation (Fig. 11, upper axes) we reversed the initial fluxes, attributing the value of 7.4×10^6 mol/year for F_m and the value of 29.45×10^6 mol/year for F_f , respectively. In the results according to models displayed in Fig. 11 upper axes it is shown that only

the magnitudes of the fluxes F_m and F_f would have changed, but there is no net change in the relative proportions between them (i.e., F_f/F_m is the same when comparing conditions between models from Fig. 11 bottom and upper axes), indicating that the same inferences as those made above are valid. With this, we show that the fluxes F_f were indeed very likely higher than F_m , irrespective of the initial (and real) values of them both, all the other parameters being constant.

The considerably distinct isotopic composition between the Jiquiá and Alagoas carbonates, from about 0.711 to 0.713 in their respective isotopic compositions, is another indication that a progressively more important source of waters to the lake should have been from groundwater recharge. A prolonged residence time within rocks possessing more radiogenic Sr values could drive the values of the lake water, and consequently of the carbonates, towards more radiogenic values too. Similarly to what occurs with groundwaters of lake basins in present-day eastern Africa (e.g., Ojiambo et al., 2003), long residence time amidst more acidic bedrock and/or detrital alluvial material eroded from basement rocks, could provide waters from which the Alagoas deposits formed with more radiogenic $^{87}\text{Sr}/^{86}\text{Sr}$ isotope composition. The balance between this contribution and that from fluvial input, which we assume had a lower $^{87}\text{Sr}/^{86}\text{Sr}$ isotope composition, seems to be what drove the evolution of the $^{87}\text{Sr}/^{86}\text{Sr}$ isotope composition of lake water.

In some regions of the pre-salt, seepage via groundwater discharge could result in travertine-like structures in fault-associated areas. Distant from fault zones, though, this recharge simply implies contribution to maintain the lake water at some given level. This is particularly the case if riverine input decreases or evaporation rate increases with time, reflecting intensification of a drier climate, which would result in greater concentration of Sr in solution. Notwithstanding, precipitation of carbonates very likely occurred as a result of concentration of the ions during increasing evaporation of lake water, because significant fluid delivery from surface runoff would result in diluted waters. This resulted in progressively more concentrated and alkaline lake water brines. This reasoning agrees with the geological observation and with the geochemical data presented earlier. Thus, the data apparently reflect a change in tectonic regime in a broader scale, with subordinated climatic control, greatly conditioned by the tectonic evolution of the basin.

Therefore, climatic conditions certainly have had an important role in the deposition of the pre-salt carbonates as a whole, but it is rather clear that these conditions were subjected to the control of the tectonic evolution of the landscape. Riverine discharge, as well as subsurface influx, were also conditioned by factors such as erosion of mountains, decreasing hydraulic head and thermal flux, as discussed above. The general results of this model indicate that the pre-salt lake in Santos Basin probably evolved from a large and deep lake, possibly comparable in size to or even larger than Lake Tanganyika in eastern Africa, to a perhaps larger and shallower, considerably alkaline endorheic lake. The thermal subsidence during the sag period that followed the rift phase is conformable to the endorheic lake postulate that very likely was established thereafter (Talbot and Allen, 1996; Reston, 2010).

5. Conclusions

The present study reinforces the lacustrine origin of the pre-salt deposits from Santos Basin, and proposes a plausible model for the general conditions of their formation, suggesting that a somewhat freshwater lake was eventually replaced by a terminal highly alkaline evaporative lake. Initially apparently counterintuitive, the strontium isotope data show that the lake water evolved to more radiogenic values approaching the top of the pre-salt succession. This is evidence that the lake evolved for a long time completely disconnected from the oceans, basically controlled by the catchment areas and subsurface recharge system inputs that contributed in different proportions relative to one another over time, until the ocean finally had some influence

over the continental sedimentation. This last phase though was not recovered and cannot be seen clearly from the sampled material, as the transition itself may be completely eroded, the unconformity at the base of the salt being the only remaining feature left from a period after that.

Our mass-balance calculations show that a felsic component end-member should have been a significant source of water and ions, and of Sr, to the palaeo-lake. To achieve the radiogenic values observed in the carbonate rocks the felsic source had likely contributed with fluxes one to two, and up to four, orders of magnitude higher than the mafic source. Finally, the results obtained from the geological data integrated with the numerical models suggest that the overall higher order trend of decreasing fluxes from both sources with time is the more likely scenario, suggesting that the formation of the deposits became very likely constrained to progressively more evaporitic and alkaline lake waters.

Regarding hydrothermalism, we do not mean by considering its occurrence that it is implied that all the deposits studied here represent travertine-like, fault associated carbonate build-ups. Rather, our point is that subsurface water recharge to the lake probably occurred via conduits scattered through the lake bottom and as seepage not necessarily linked to them. In some areas fault related build-ups may have formed, but far from faulted zones these fluids simply mixed with the main body of water, significantly contributing to the chemical and isotope composition of the carbonate rocks. Although the box model is considerably simple, it agrees well with the geological observations and provides useful insights into the evolution of the rift to post rift lake system, allowing to quantitatively reconstruct a plausible scenario for the deposition of the pre-salt carbonates. Thus, it seems an appropriate model that allows the deposition of both lake bottom and fault-associated carbonates more generally.

Finally, we have worked through many uncertainties, but nonetheless we hope to have shown that our model represents a valid approach to help constrain and elucidate the scenario in which the pre-salt carbonates were deposited.

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